Project Seminar 2026





Design of a fuel cell monitoring board in KiCad and development of the operational board software

Lesson 9: Discussions on PCB designs (Part 1)

M. Eng. Andreas Hein

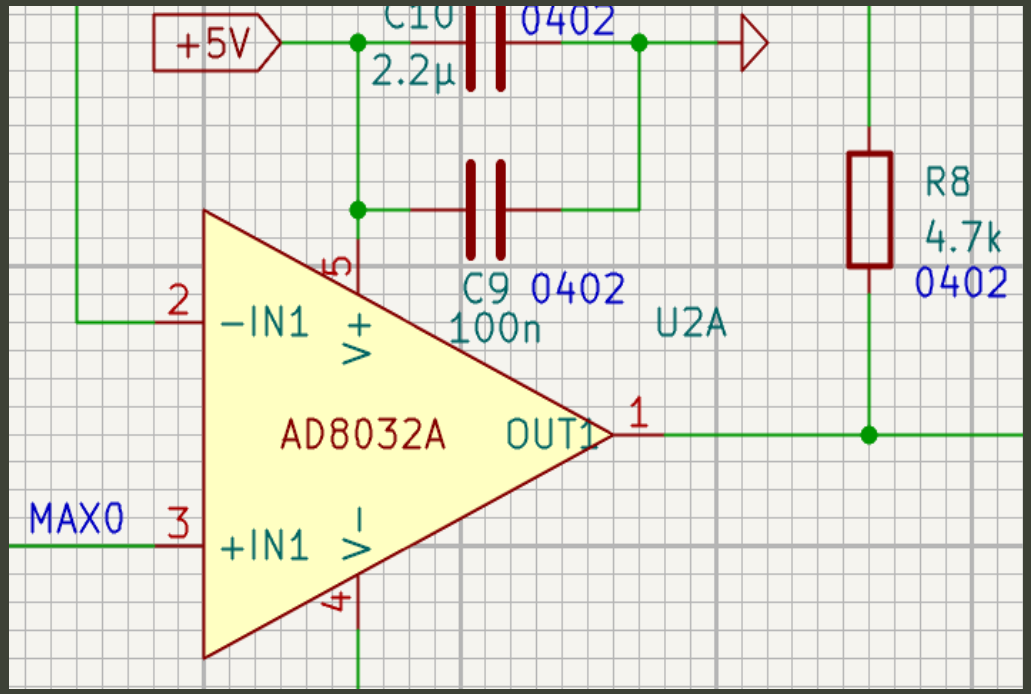
E-Mail: andreas.hein@hs-wismar.de

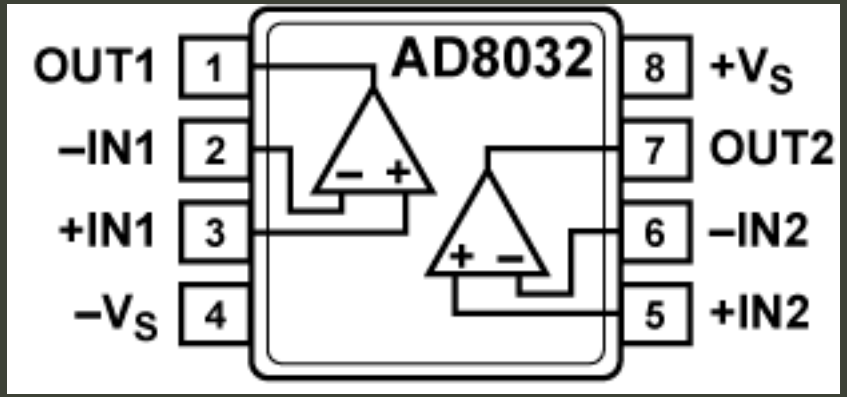
FIW: Fachbereich Elektrotechnik und Informatik

HOCHSCHULE WISMAR
UNIVERSITY OF APPLIED SCIENCES TECHNOLOGY, BUSINESS AND DESIGN

Project Seminar – PCB design

- ***AD8032 → Symbol error:***





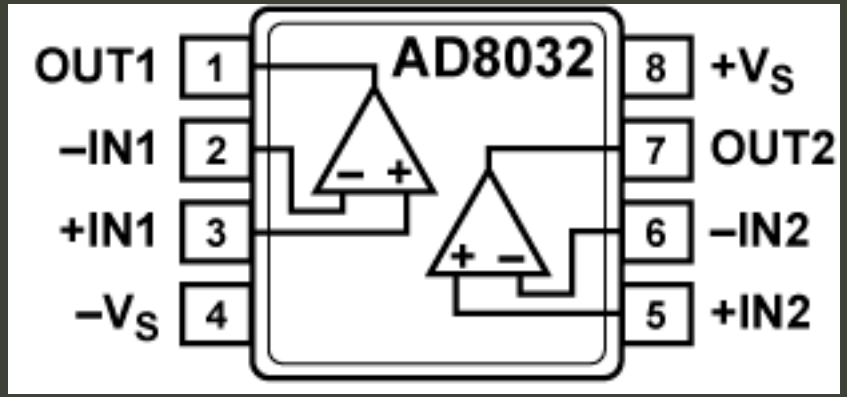
Project Seminar – PCB design

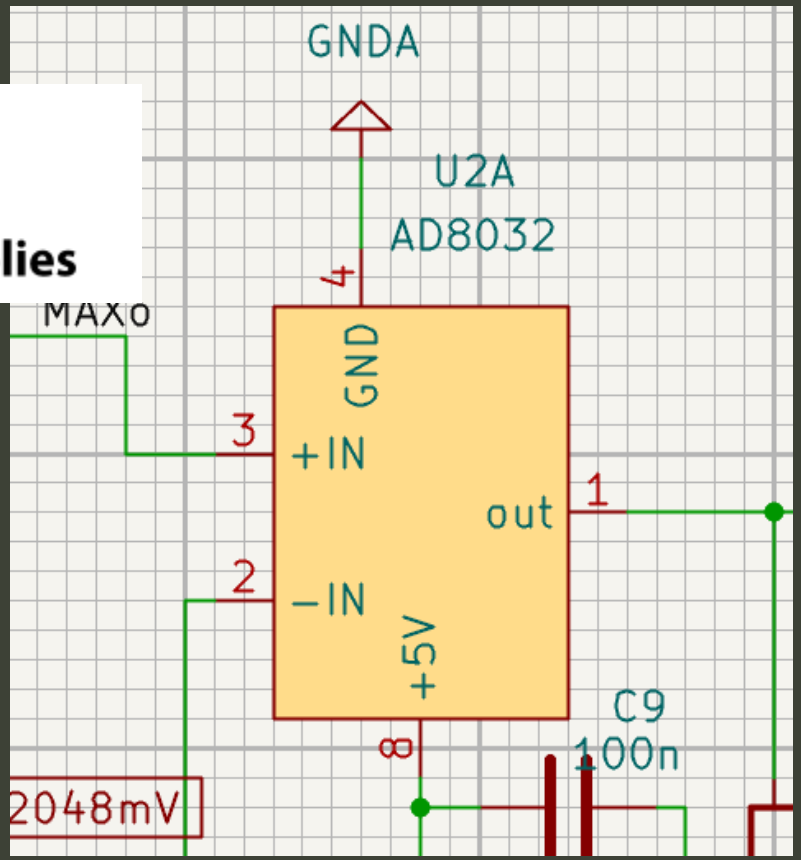
- ***AD8032 → Symbol error:***

Low power

Supply current 800 µA/amplifier

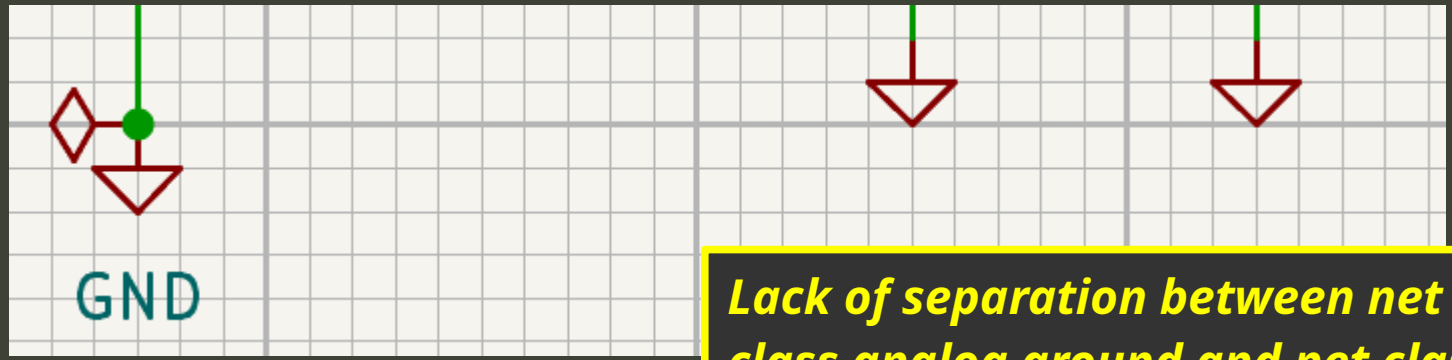
Fully specified at +2.7 V, +5 V, and ±5 V supplies





Project Seminar – PCB design

- ***Net class error:***

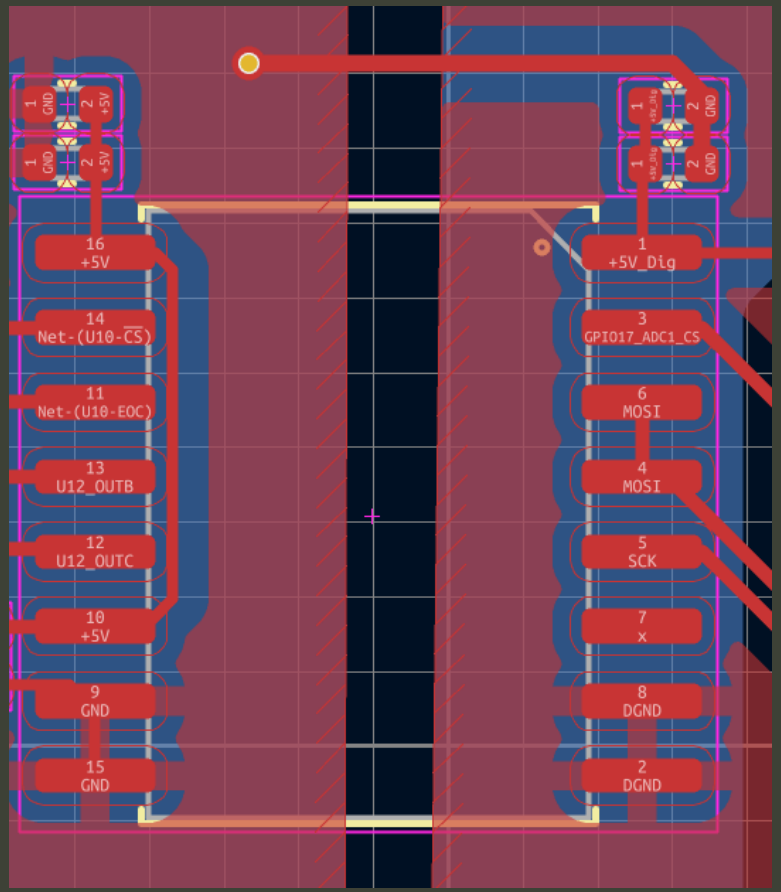


Lack of separation between net class analog ground and net class digital ground

- ***GNDA***
- ***GNDD***

Project Seminar – PCB design

- ***Footprint error:***

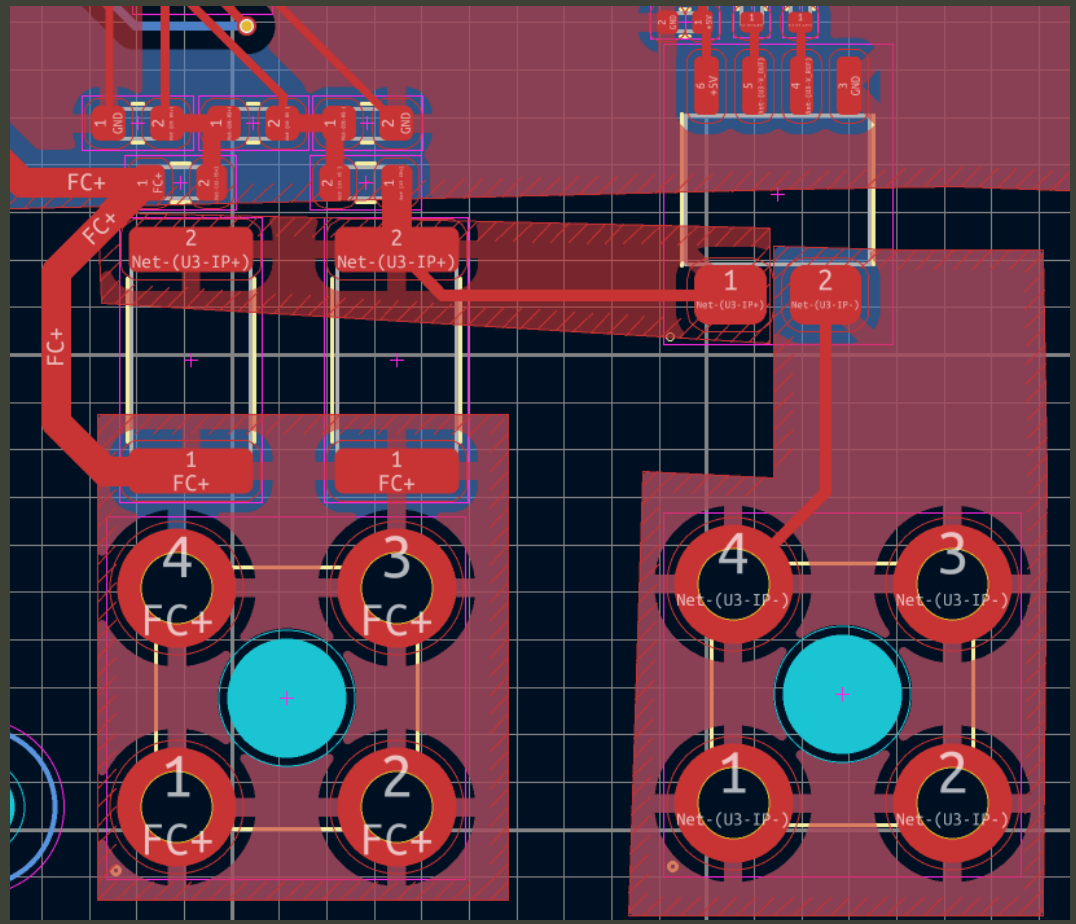


Displacement of pads

- ***Wrong pad order***

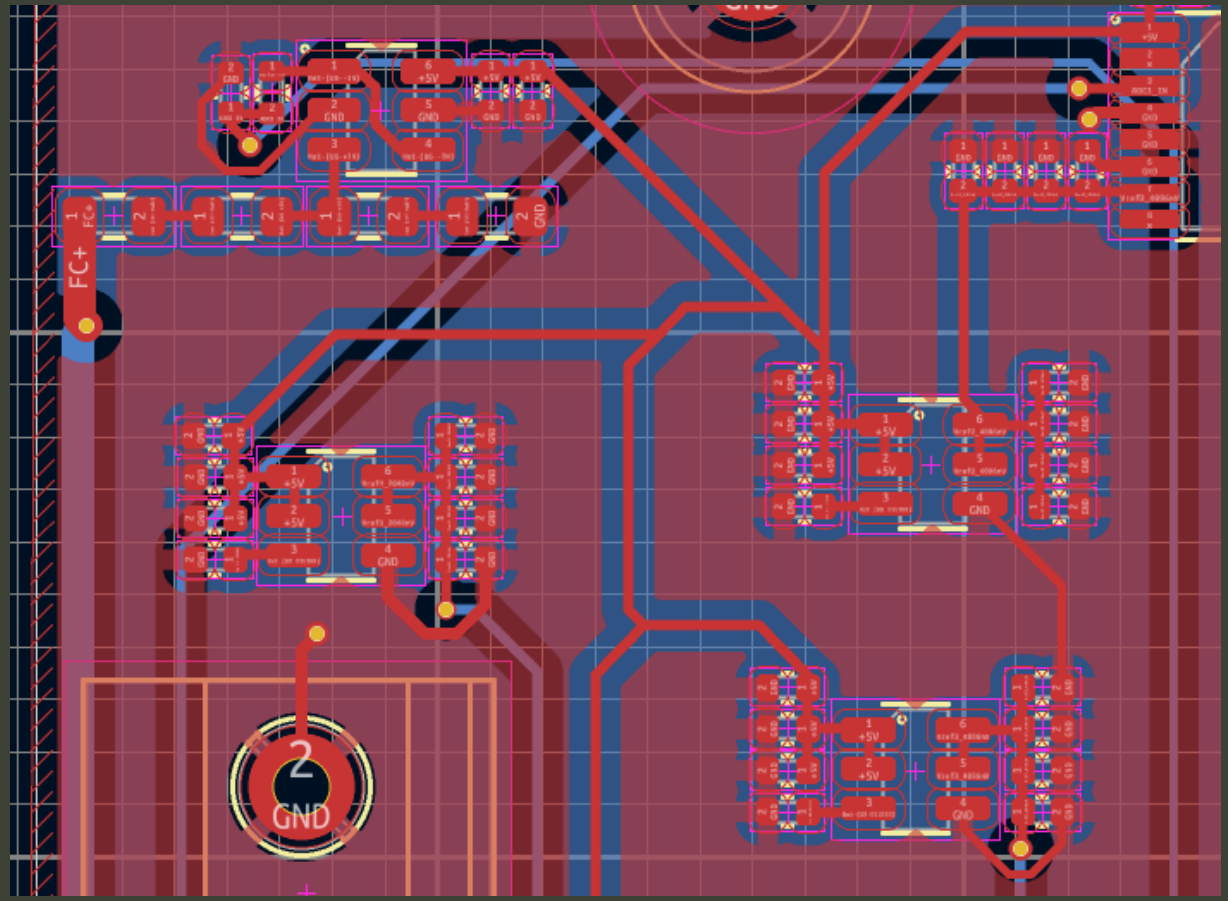
Project Seminar – PCB design

- ***Polygon design error:***



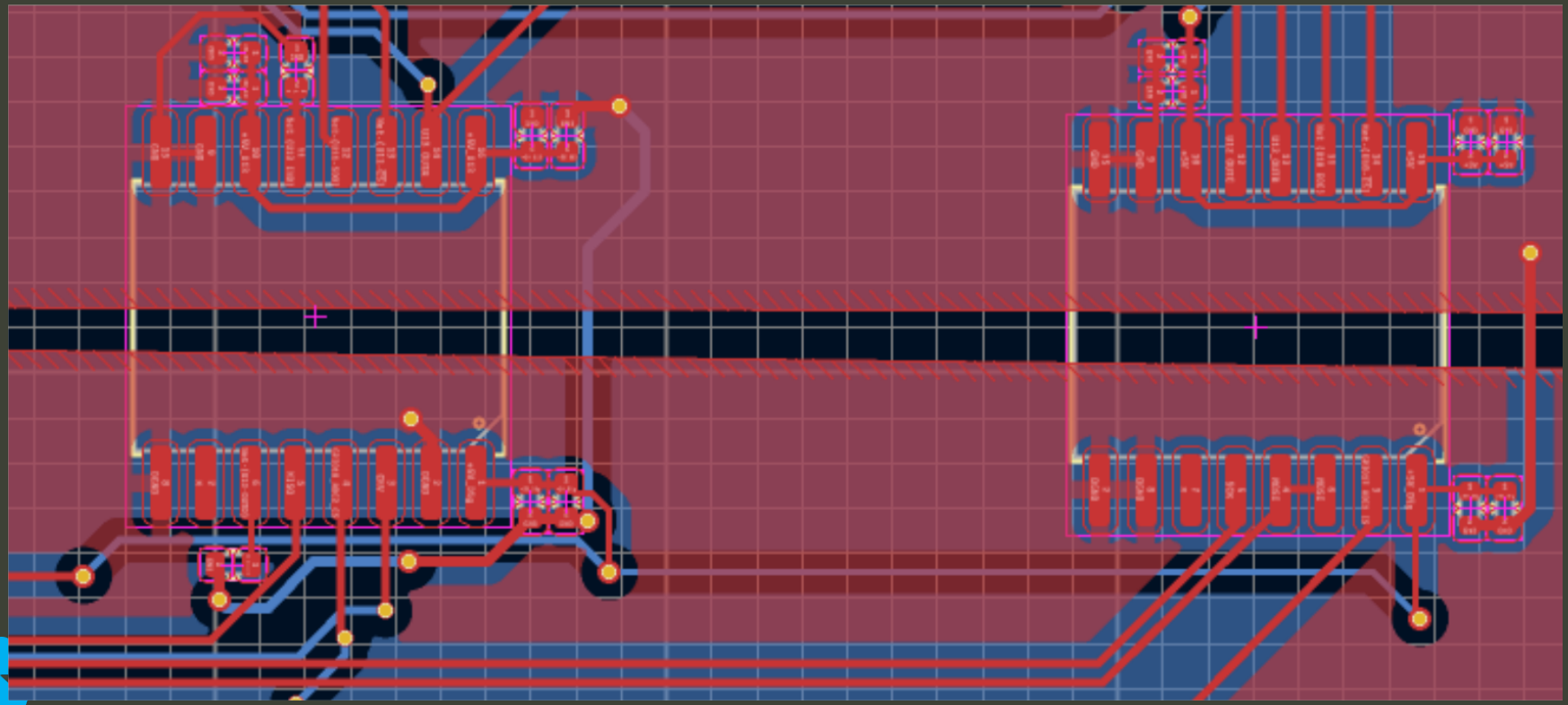
Project Seminar – PCB design

- ***Component placement & routing error:***



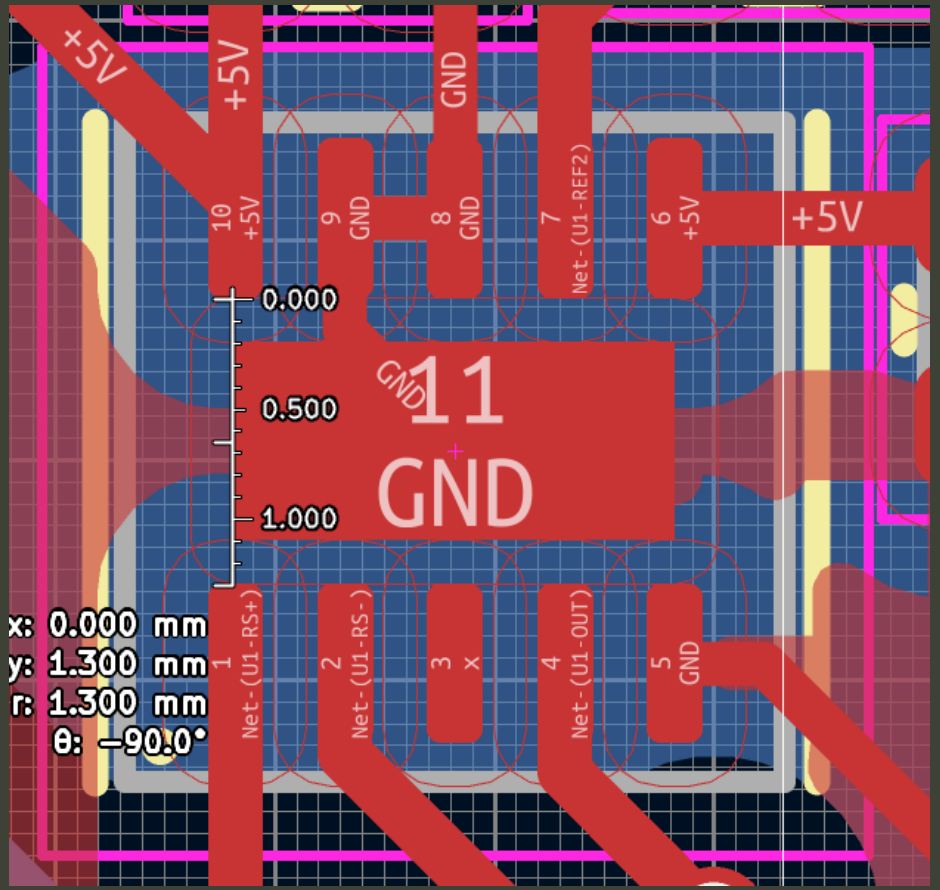
Project Seminar – PCB design

- ***Routing error:***



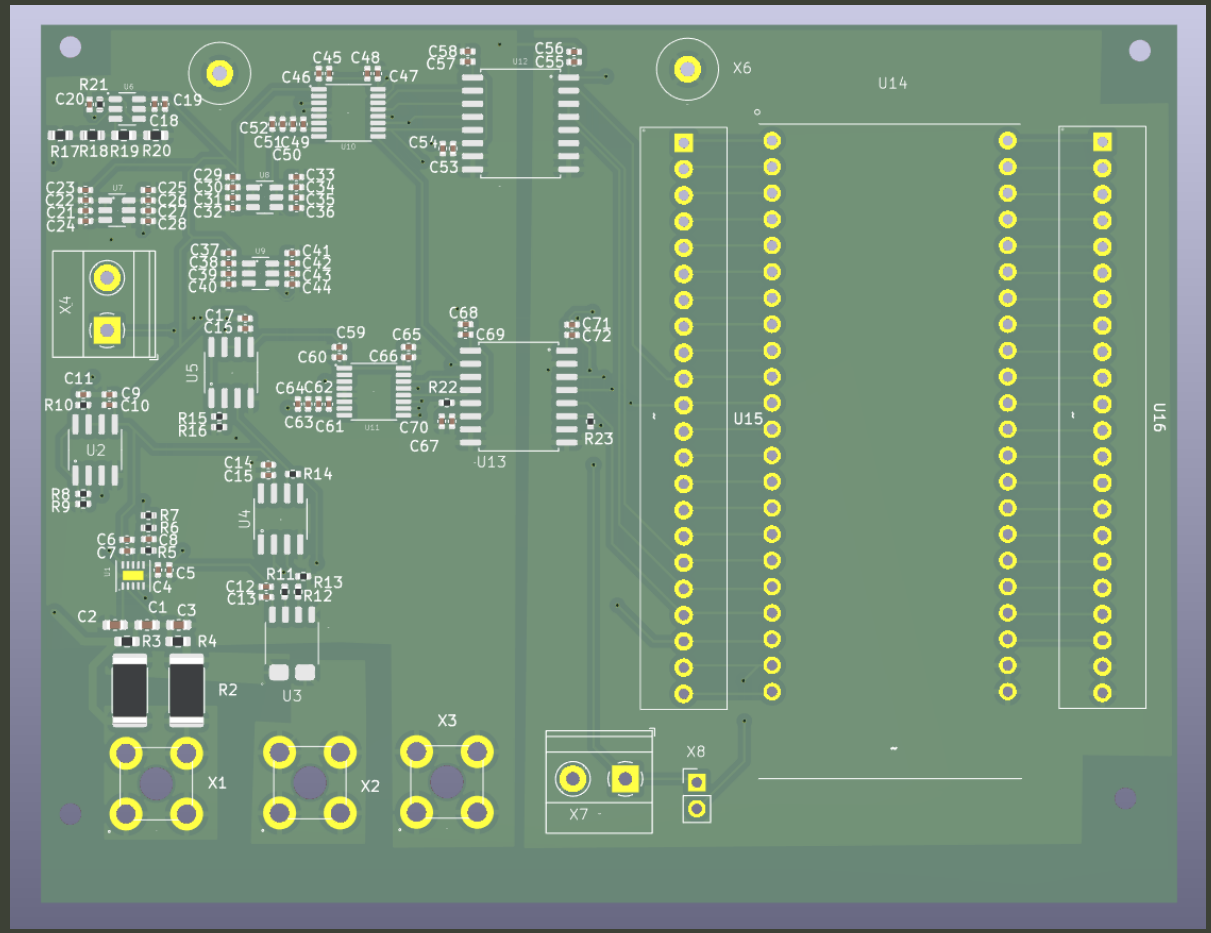
Project Seminar – PCB design

- ***Footprint (dimensions) error:***



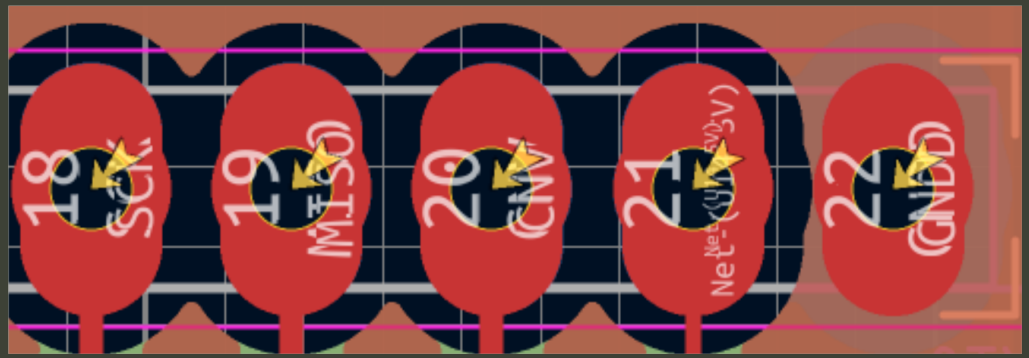
Project Seminar – PCB design

- ***Component placement error:***



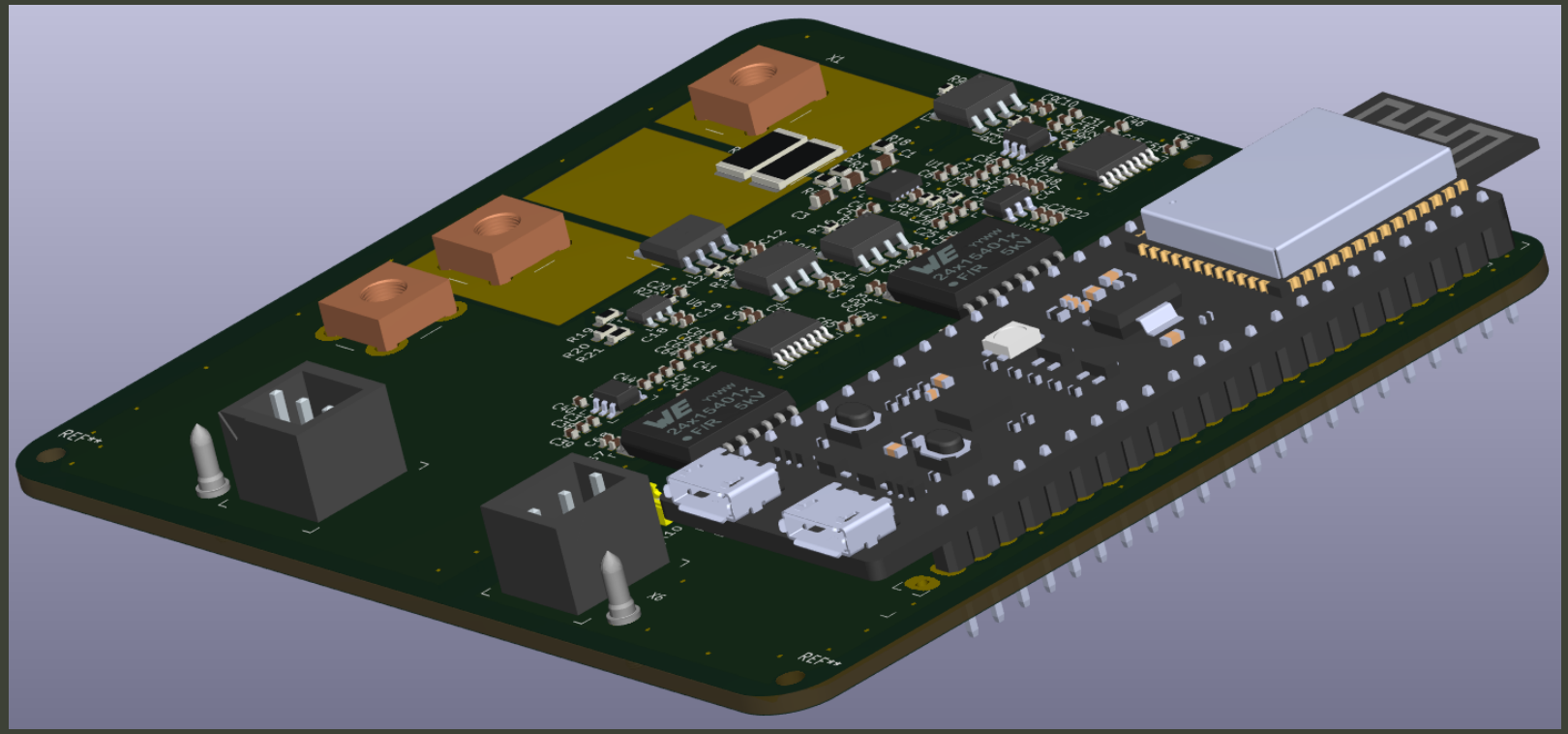
Project Seminar – PCB design

- ***Clearance error:***



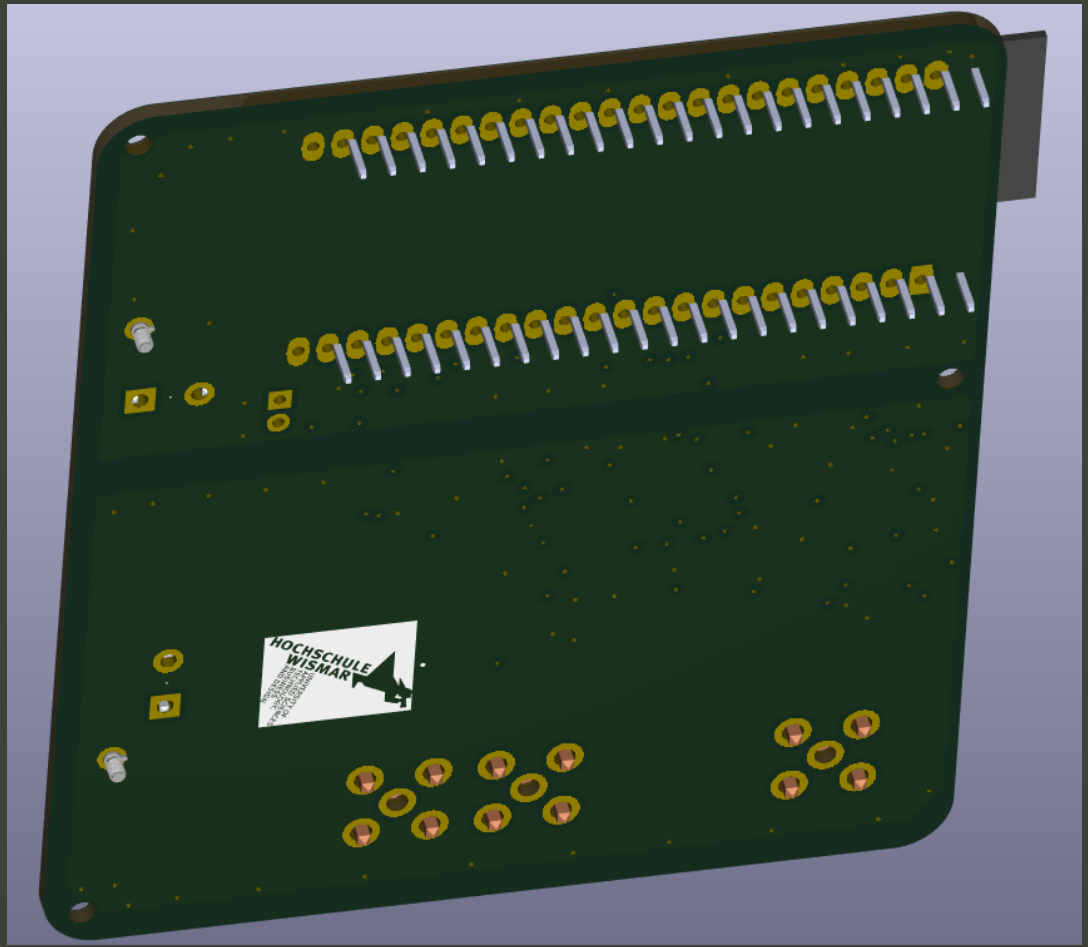
Project Seminar – PCB design

- ***3D-Model placement error:***



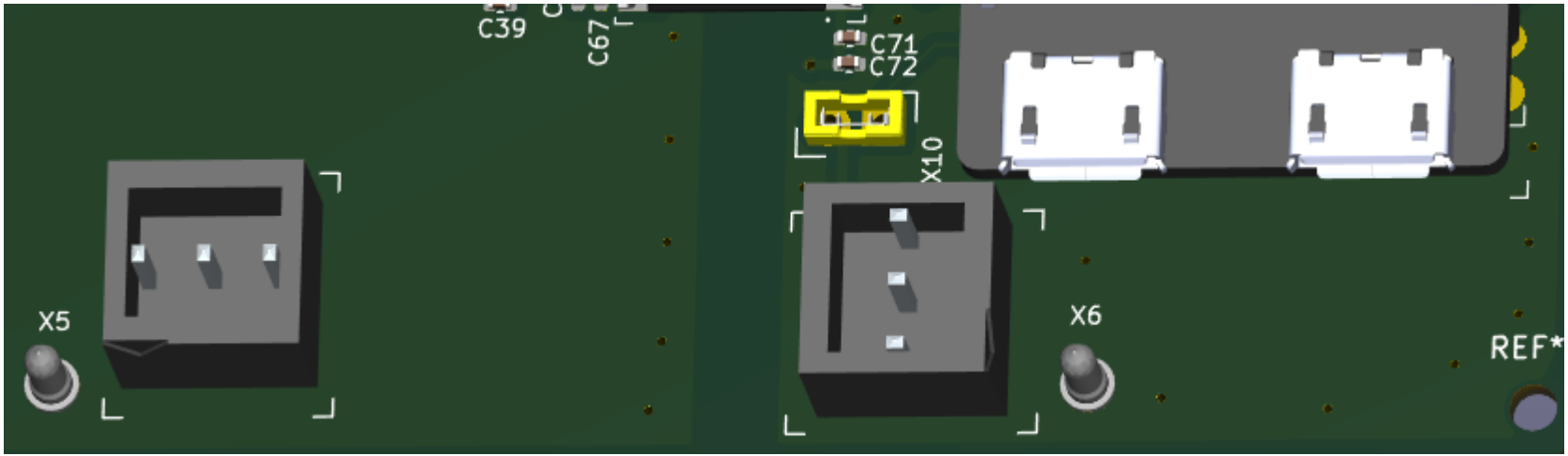
Project Seminar – PCB design

- ***3D-Model placement error:***



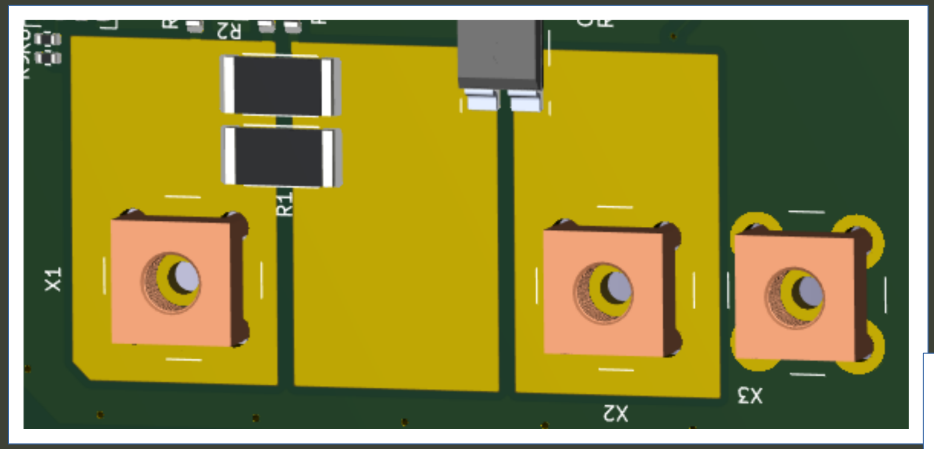
Project Seminar – PCB design

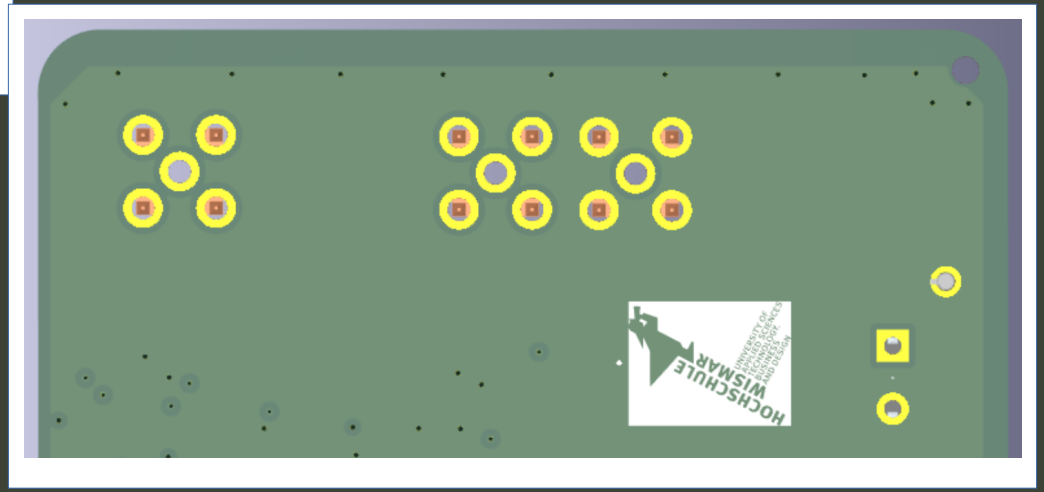
- ***3D-Modell assignment error & Orientation error & Clearance error:***



Project Seminar – PCB design

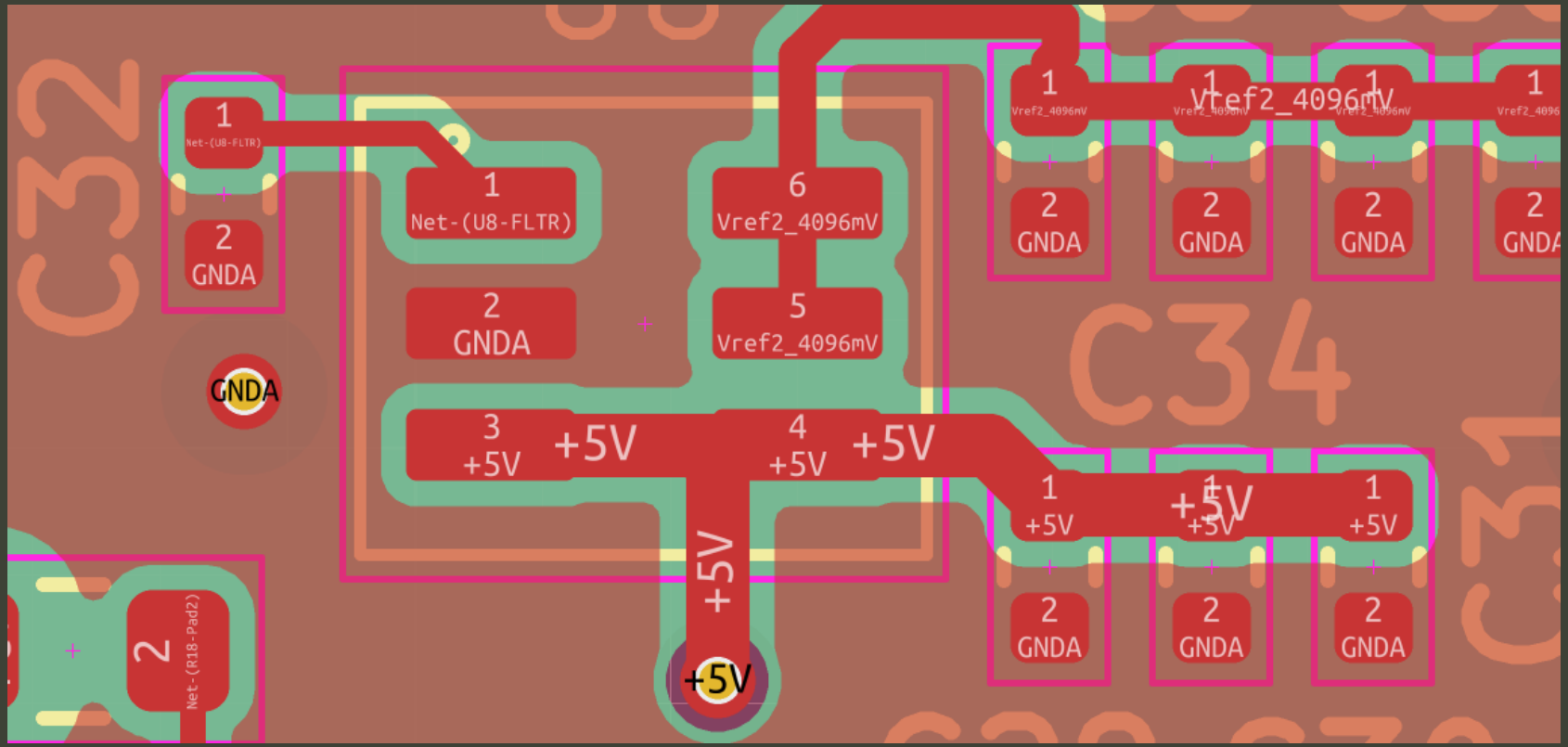
- ***Thermal management error:***





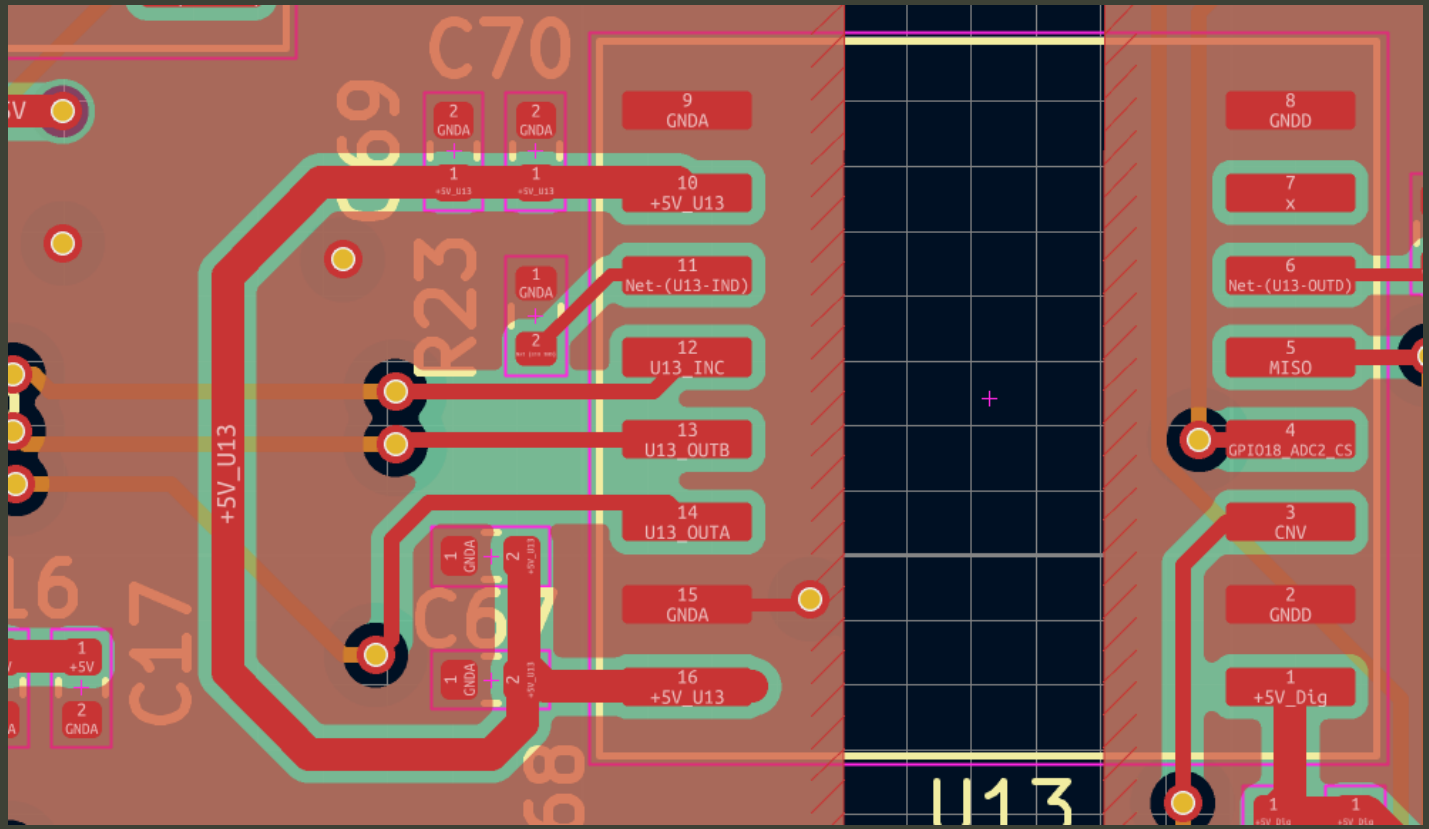
Project Seminar – PCB design

- ***Polygon error & routing error & not optimized routing:***



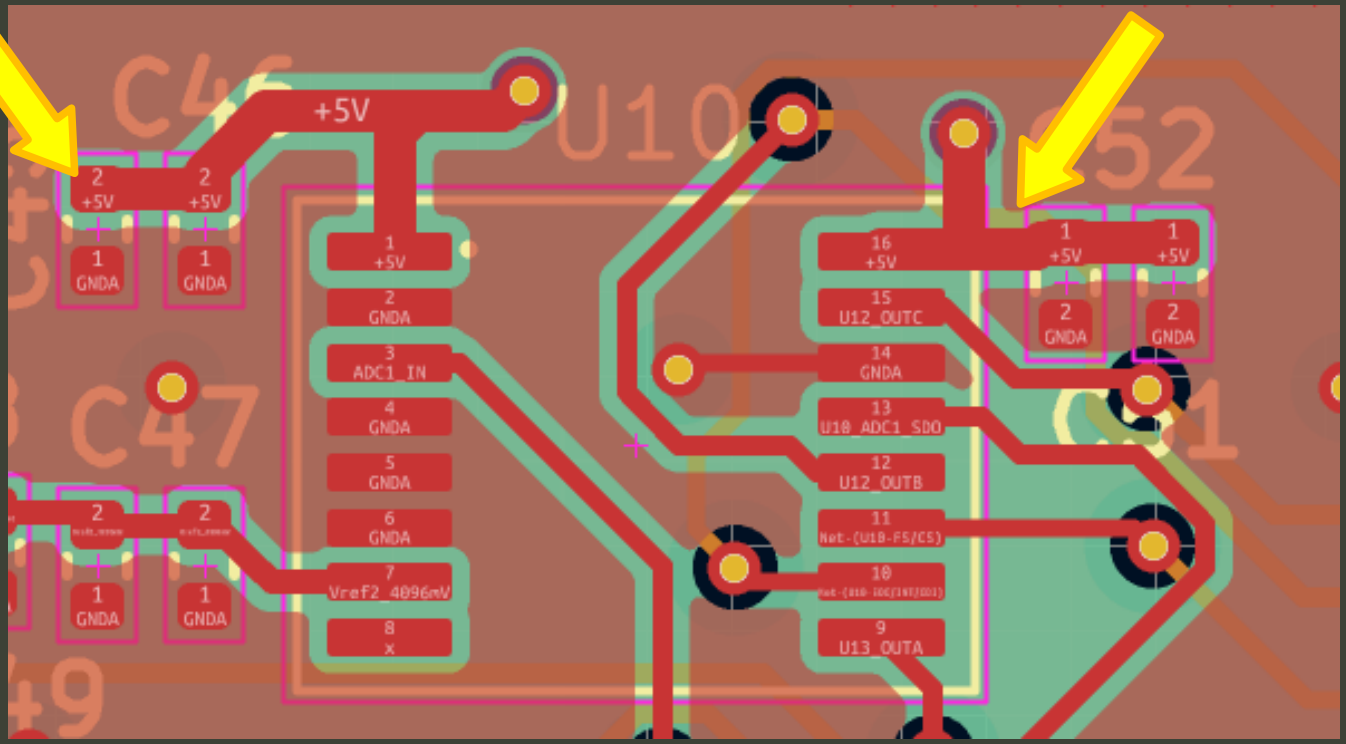
Project Seminar – PCB design

- ***Polygon error & not optimized routing:***



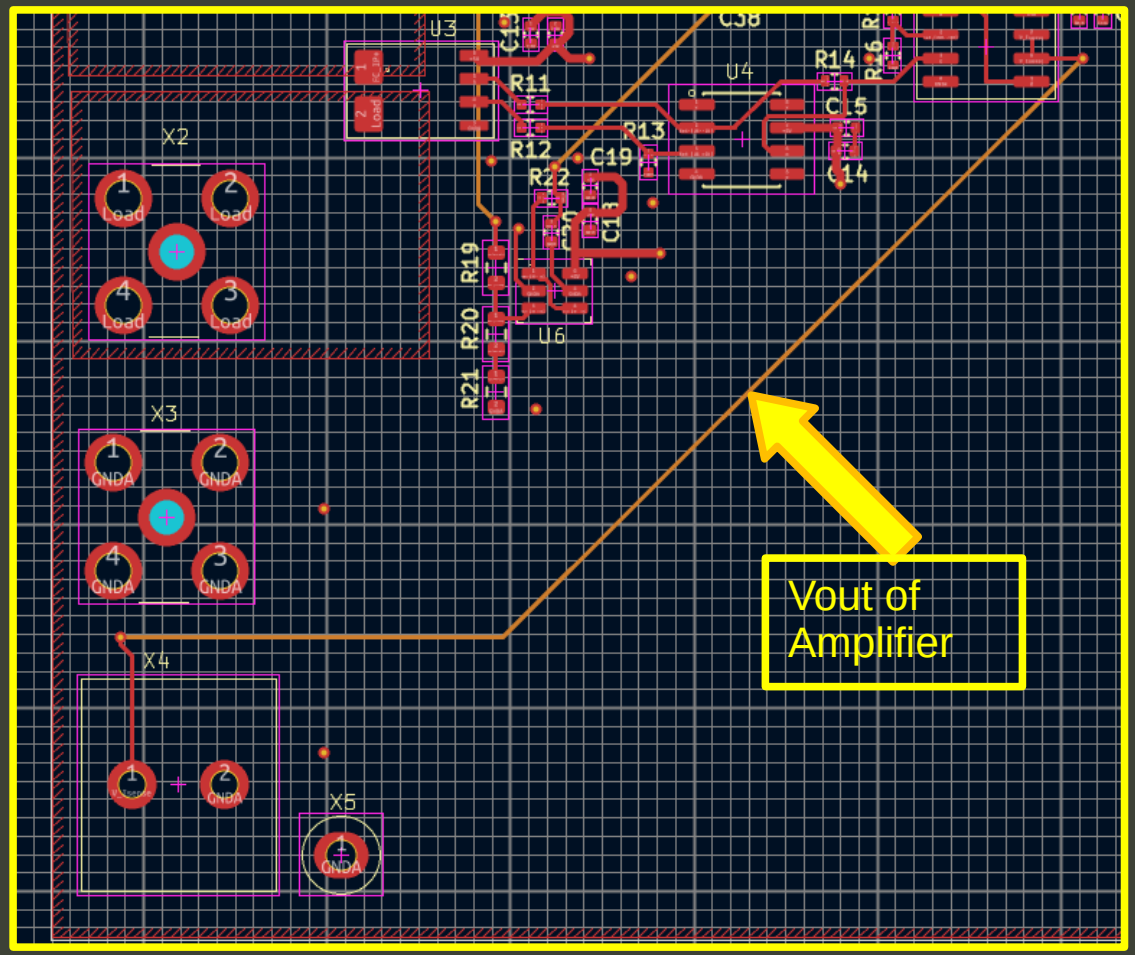
Project Seminar – PCB design

- ***Routing & placement error:***



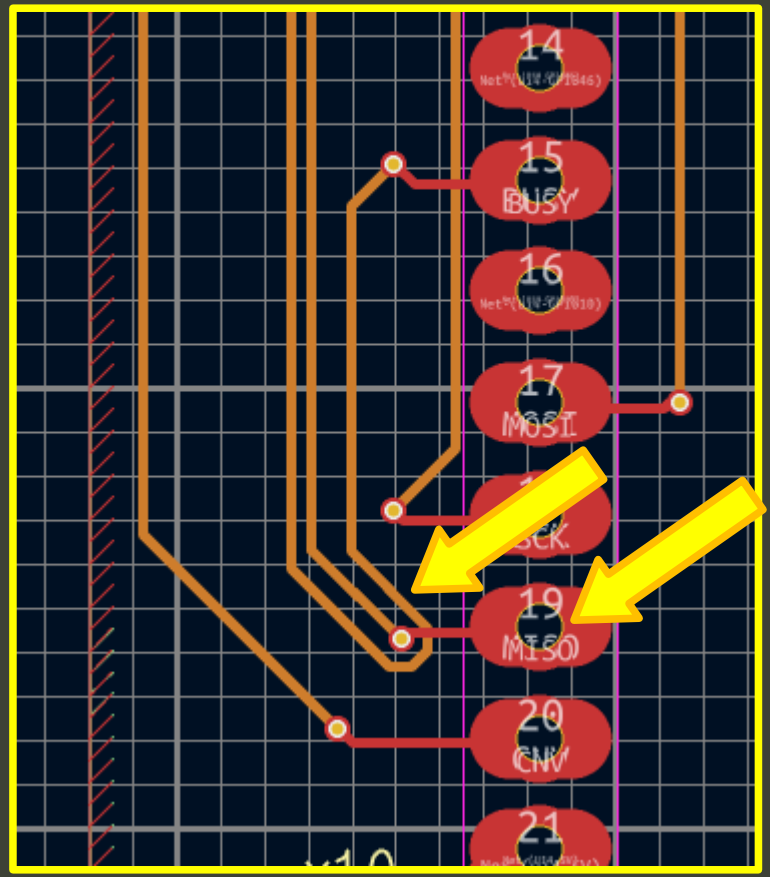
Project Seminar – PCB design

- ***Routing error:***



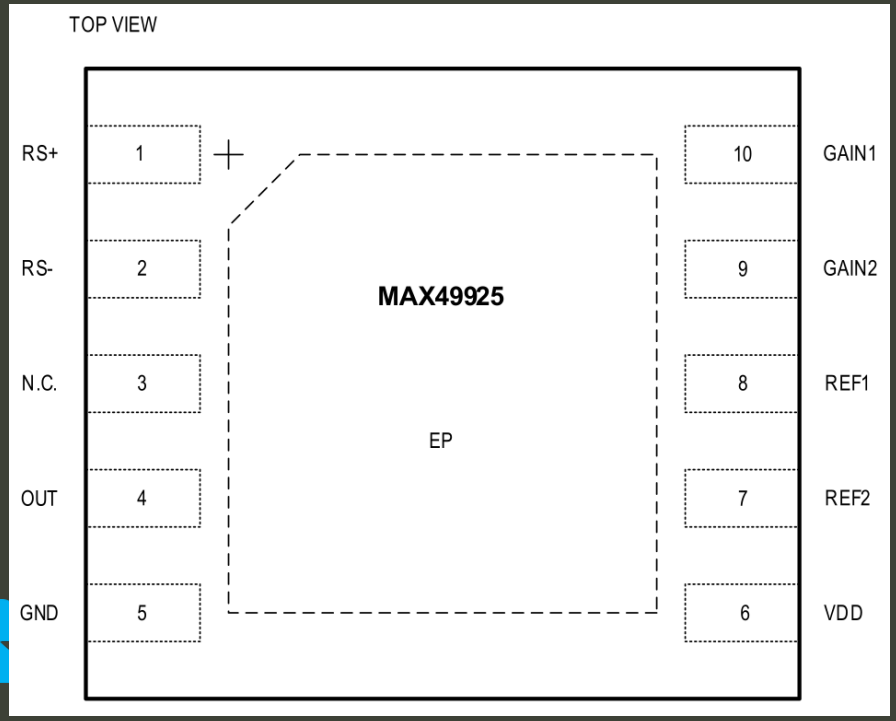
Project Seminar – PCB design

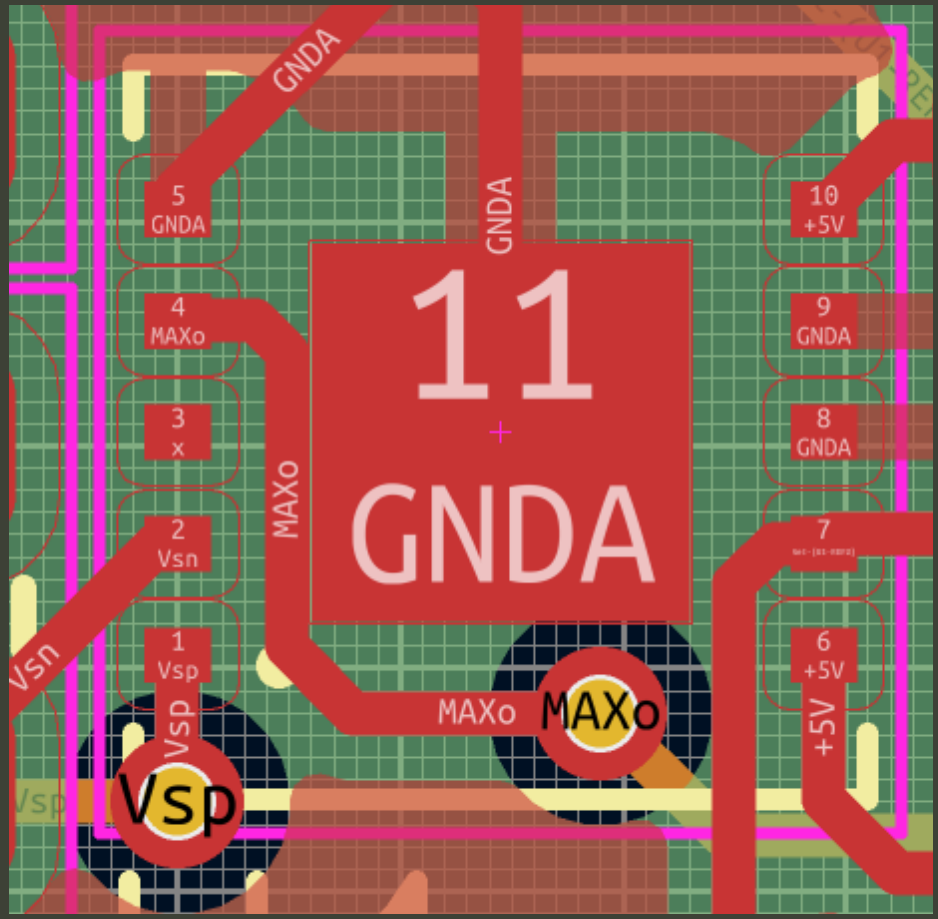
- ***Routing & clearance error:***



Project Seminar – PCB design

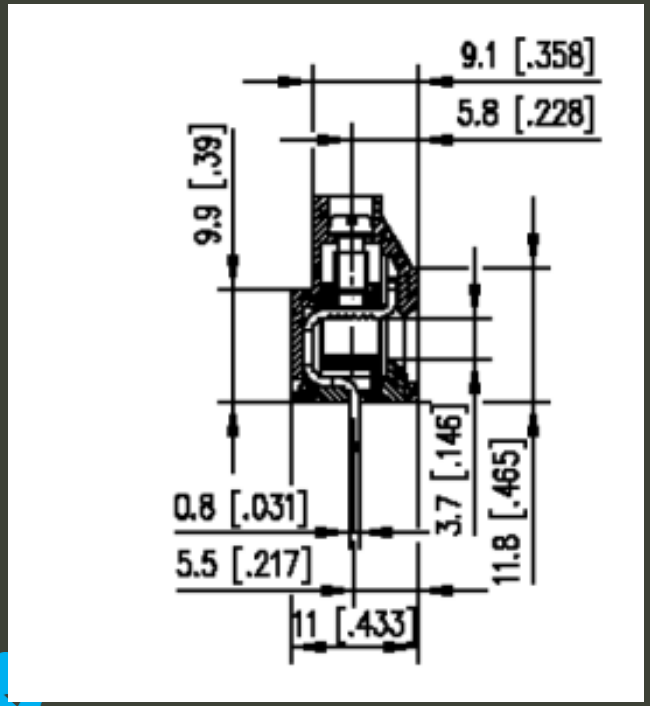
- ***Footprint & Clearance error:***

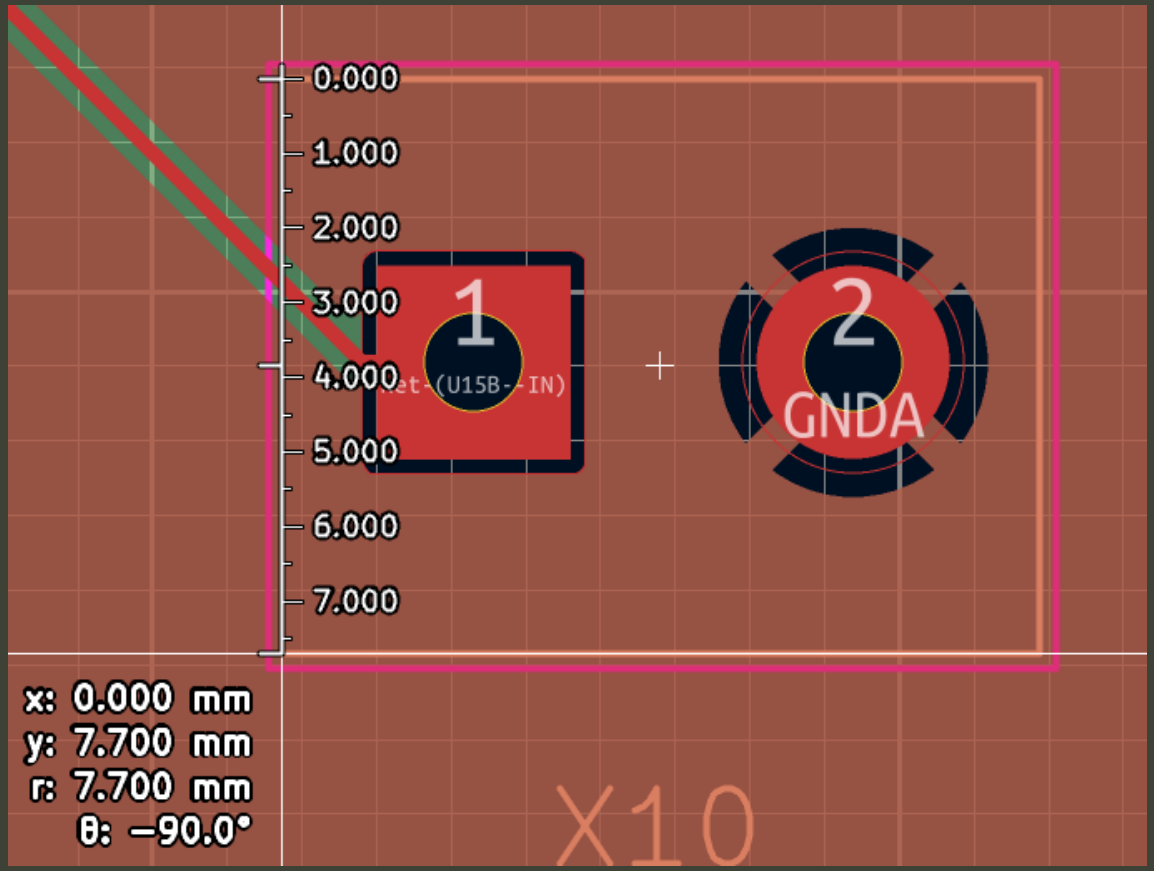




Project Seminar – PCB design

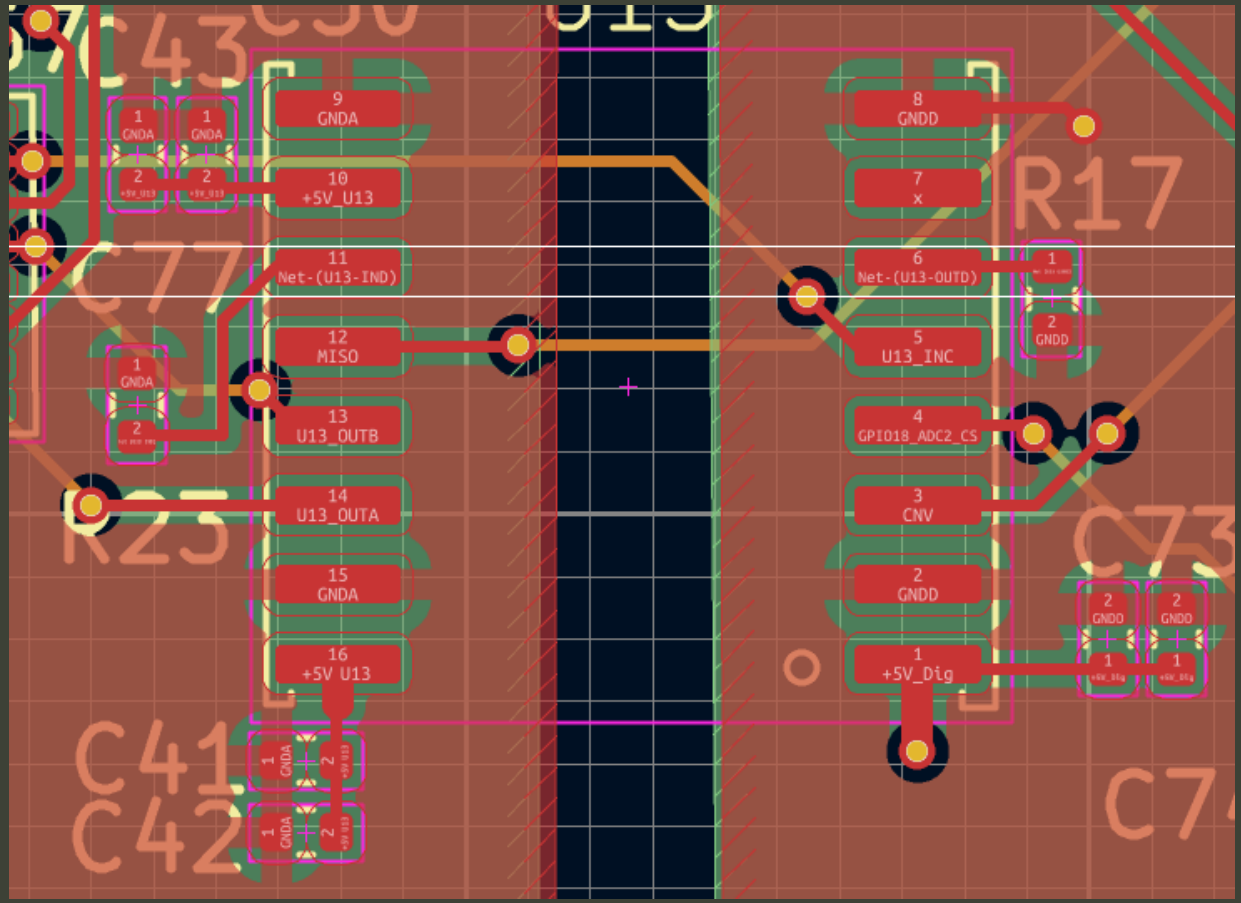
- ***Footprint error:***





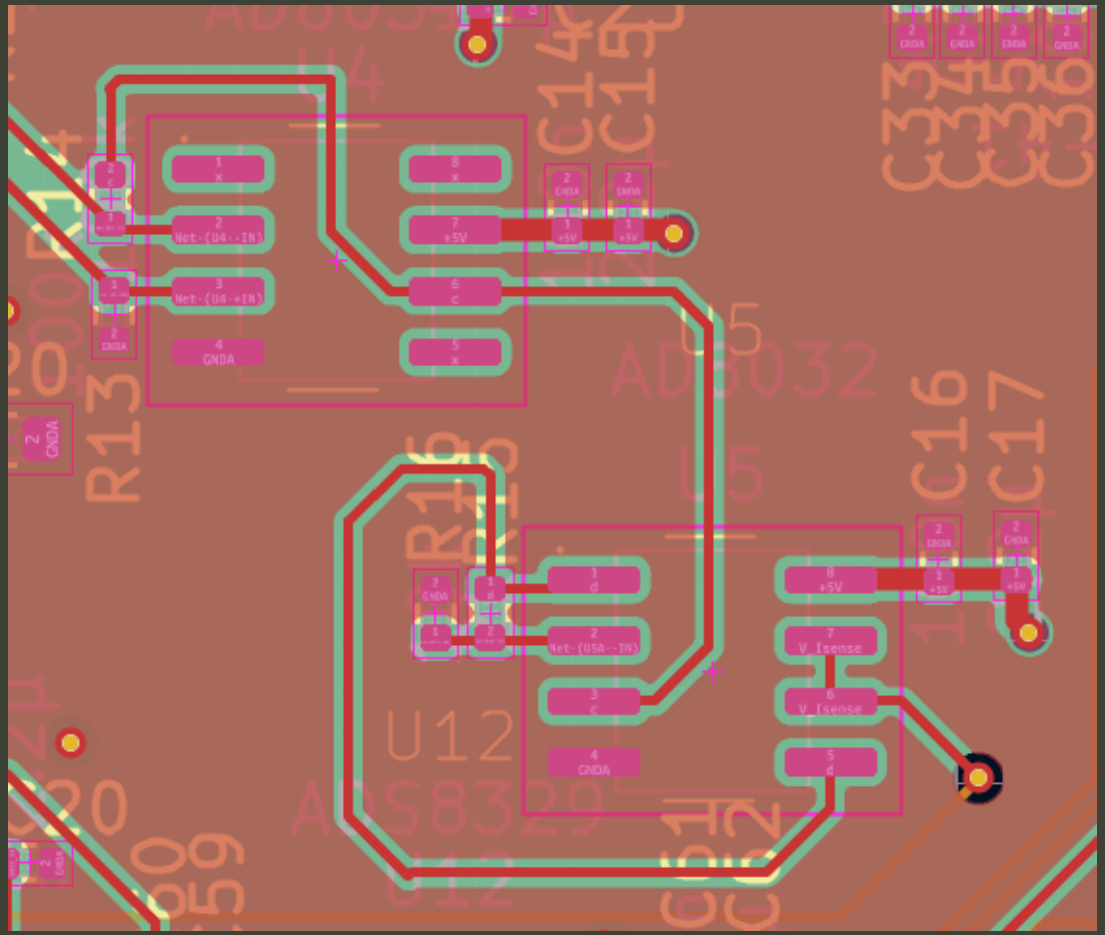
Project Seminar – PCB design

- ***Routing error:***



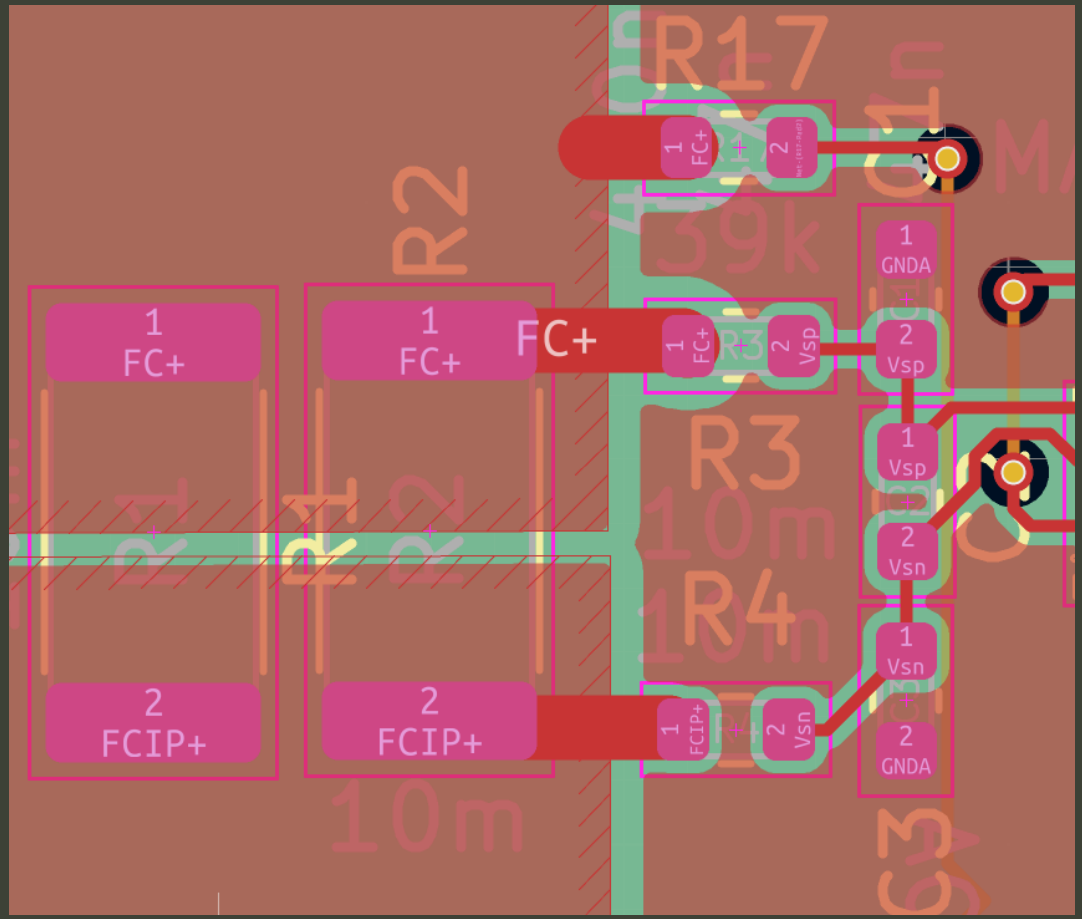
Project Seminar – PCB design

- ***Routing error:***



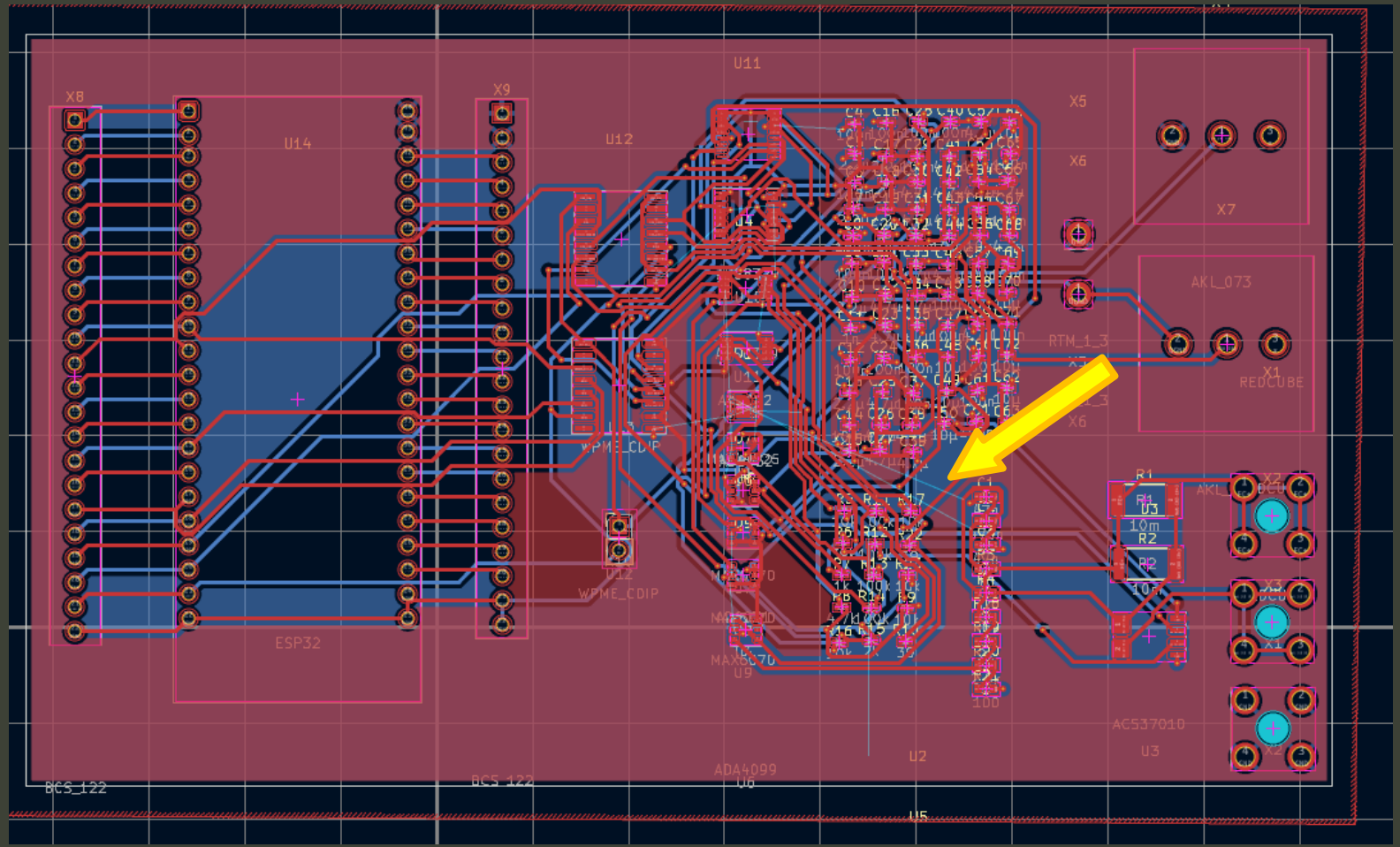
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- ***Design error:***



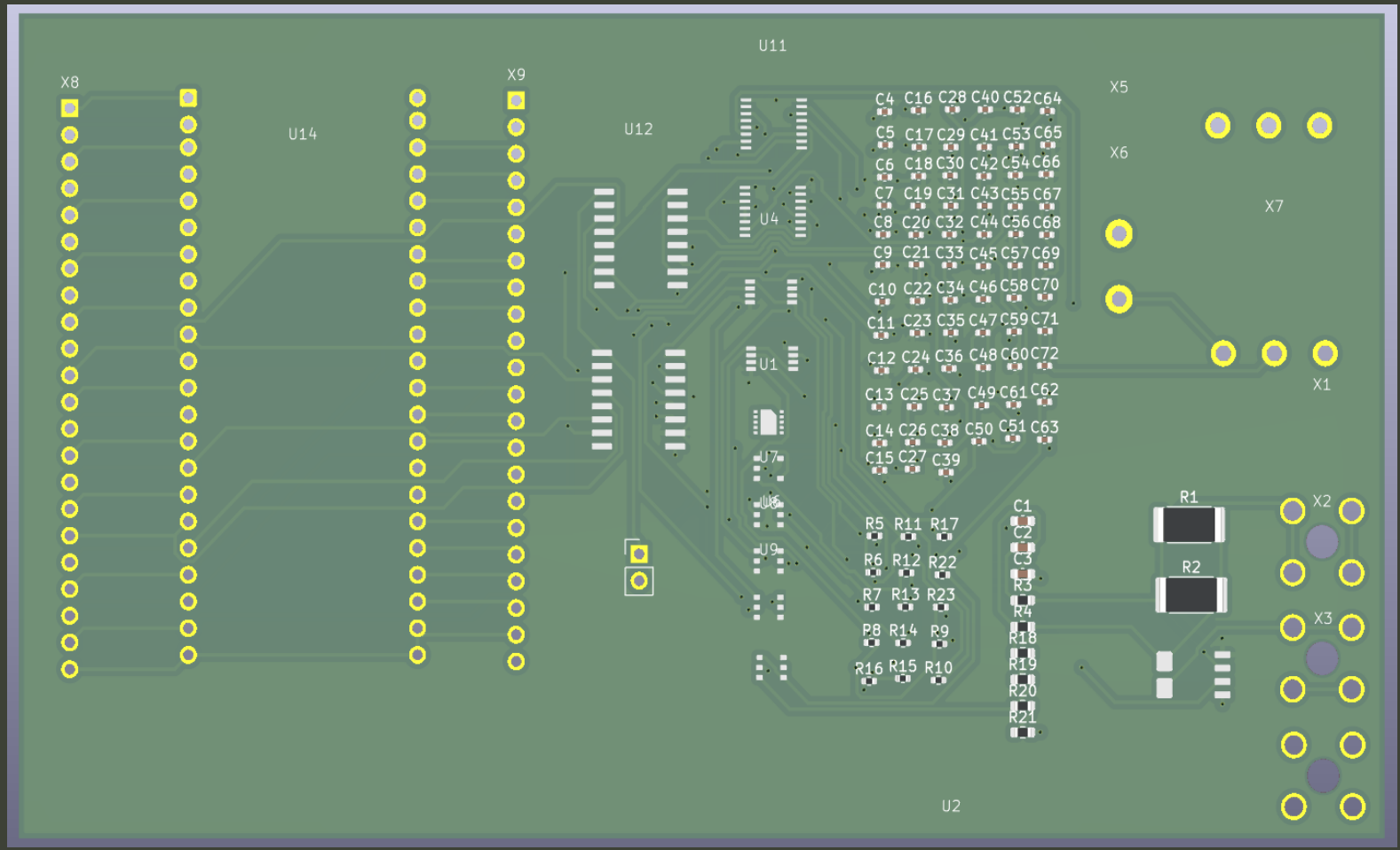
Project Seminar – PCB design

- ***Multiple fatal design errors:***



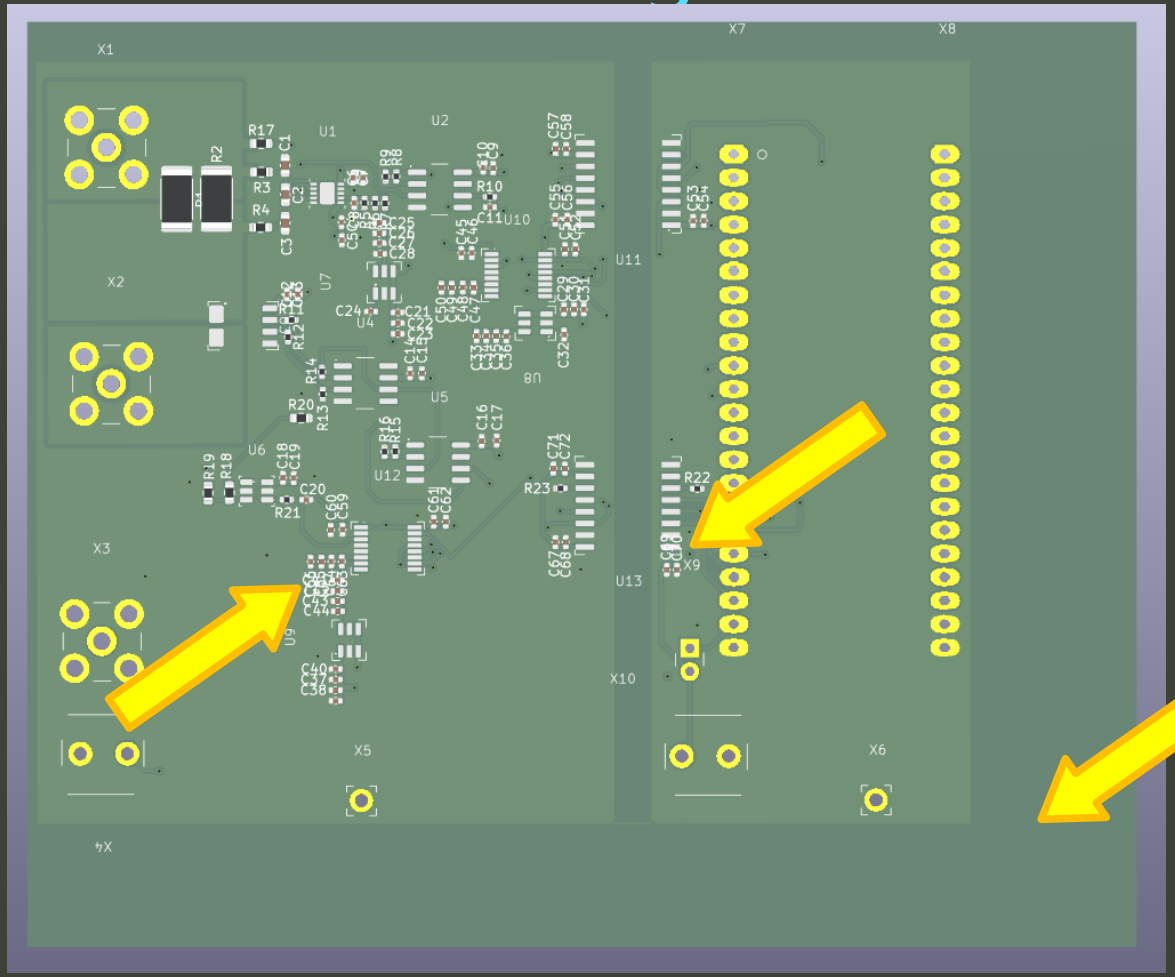
Project Seminar – PCB design

- ***Multiple fatal design errors:***



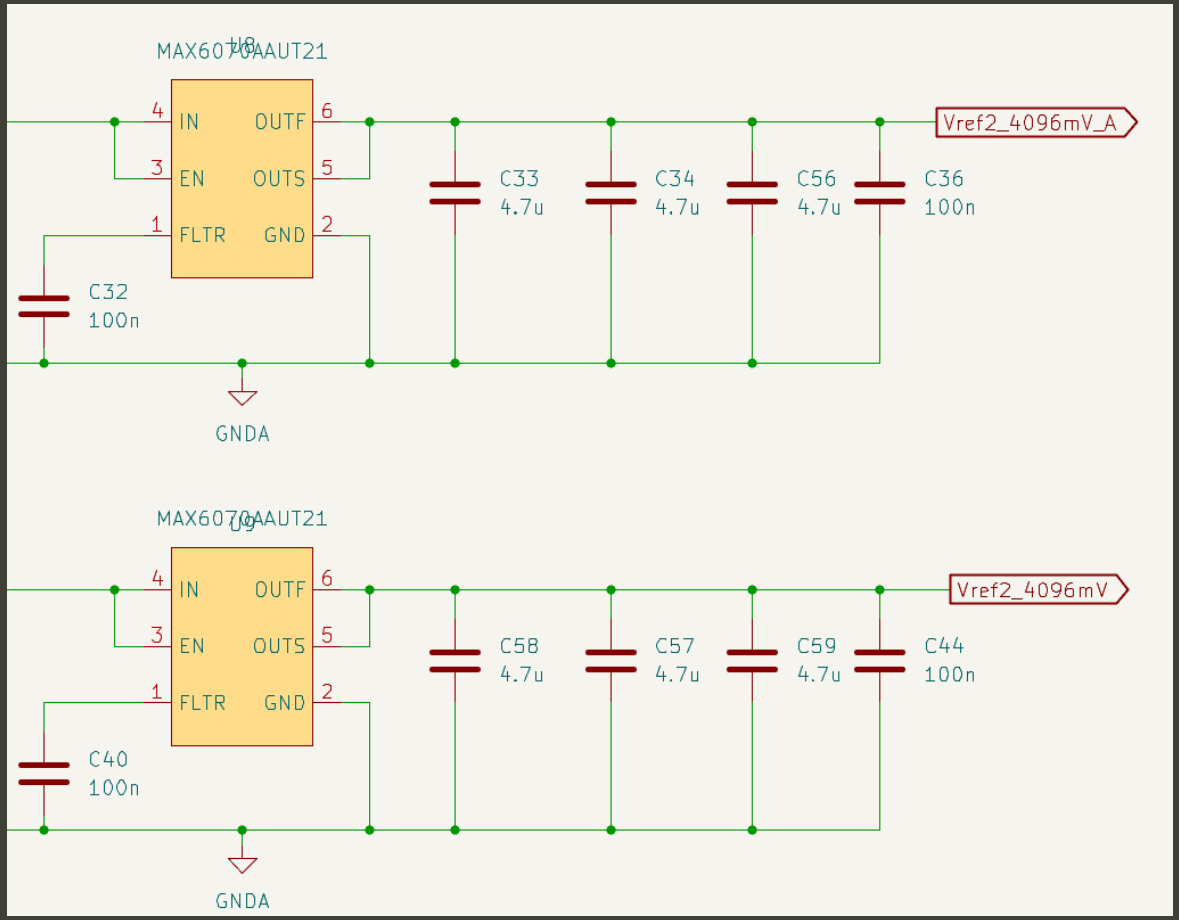
Project Seminar – PCB design

- ***Silkscreen & Edge Cut error:***



Project Seminar – PCB design

- ***Schematic error:***



Project Seminar 2026





Design of a fuel cell monitoring board in KiCad and development of the operational board software

Lesson 9: Development Operational Board Software

(Part 2)

M. Eng. Andreas Hein

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FIW: Fachbereich Elektrotechnik und Informatik

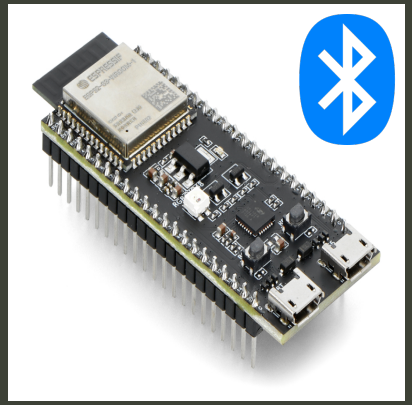
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Project Seminar – Board Software Dev.

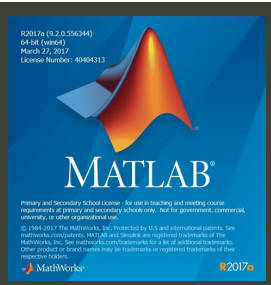
- ***Fuel cell monitoring and measurement board setup:***







GNU Octave



Fuel cell (500W)

Measurement board with ESP32-S3 micro-controller and DAQ-board software

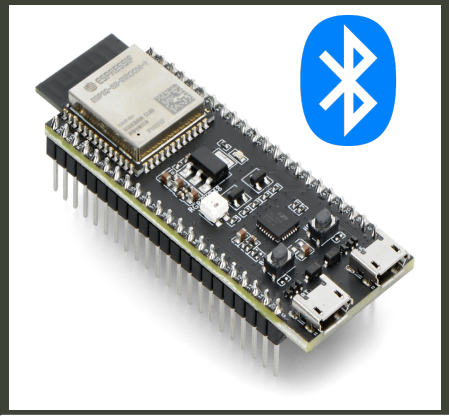
DAQ - Software

Project Seminar – Board Software Dev.

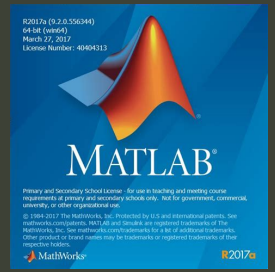
- ***System measurement chain (1):***







GNU Octave



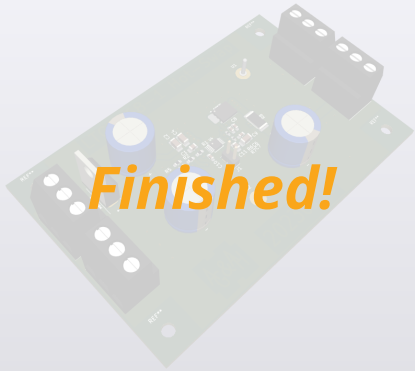
Physical quantities to be measured	Signal preprocessing and signal conditioning	Micro-Controller	DAQ - Software
→ Output current, → Output voltage	→ Sensing → Amplification → Digitization	→ Data Acquisition → Data Processing → Data Transmission	

Project Seminar – Board Software Dev.

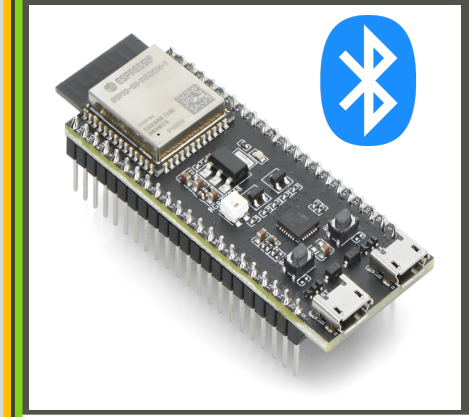
- ***System measurement chain:***



Physical quantities to be measured
→ Output current,
→ Output voltage



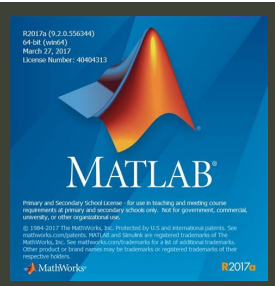
Signal preprocessing and signal conditioning
→ Sensing
→ Amplification
→ Digitization



Micro-Controller
→ Data Acquisition
→ Data Processing
→ Data Transmission

→ ***Open***

GNU Octave

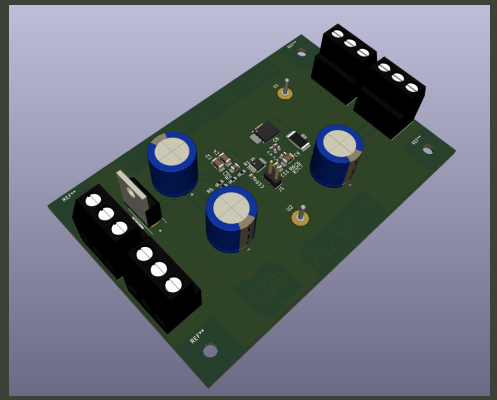


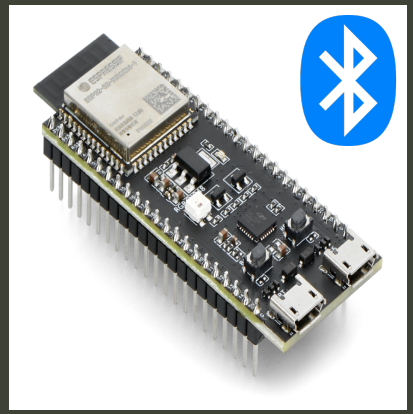
DAQ - Software

Project Seminar – Board Software Dev.

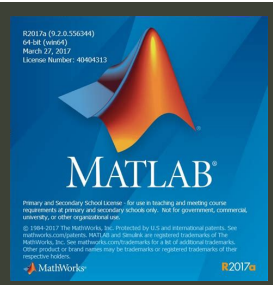
- ***System measurement chain (2):***







GNU Octave



Device Under Test (DUT) → Hardware

→

Analog to Digital Conversion → Hardware

→

ESP32-S3

→ Software

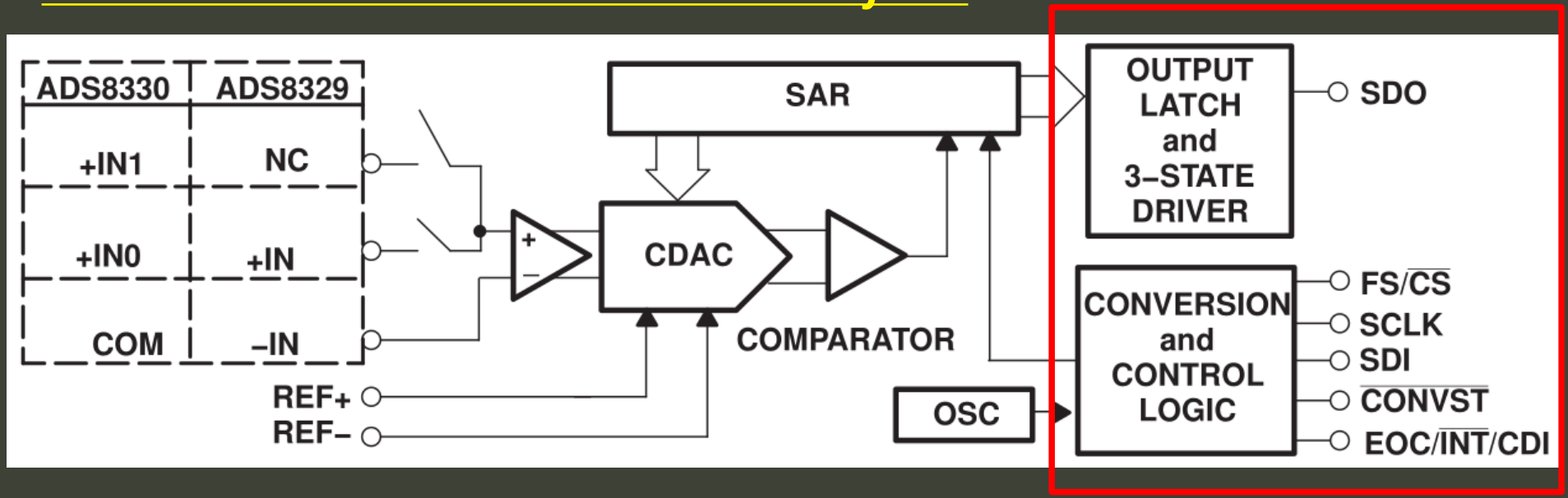
C/C++

→

DAQ - Software

Project Seminar – Board Software Dev.

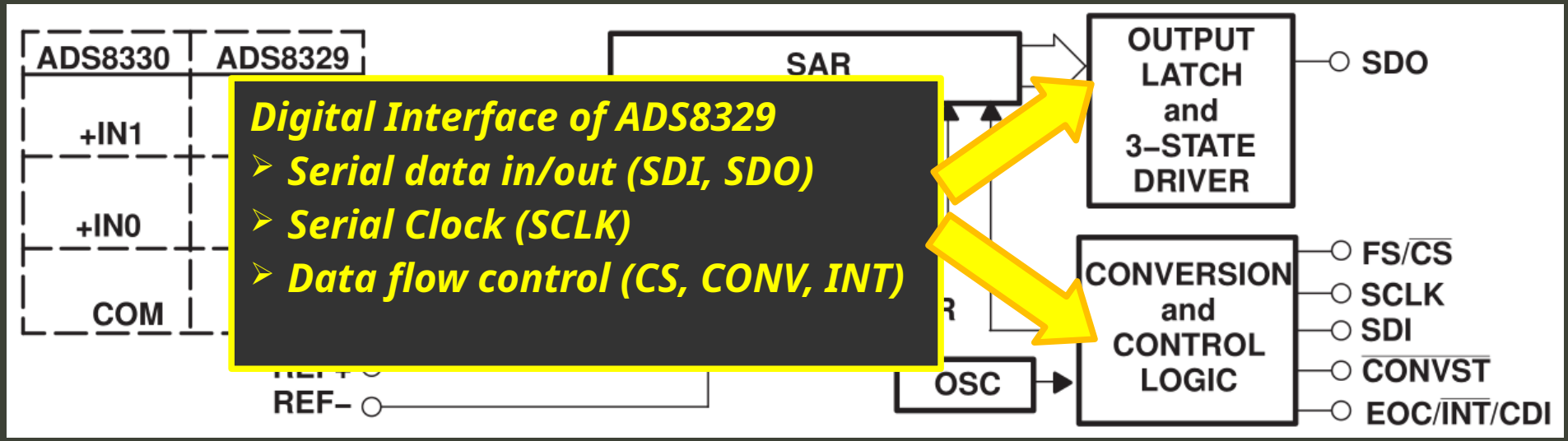
- ***ADS8329 → ADC → Communication Interface:***



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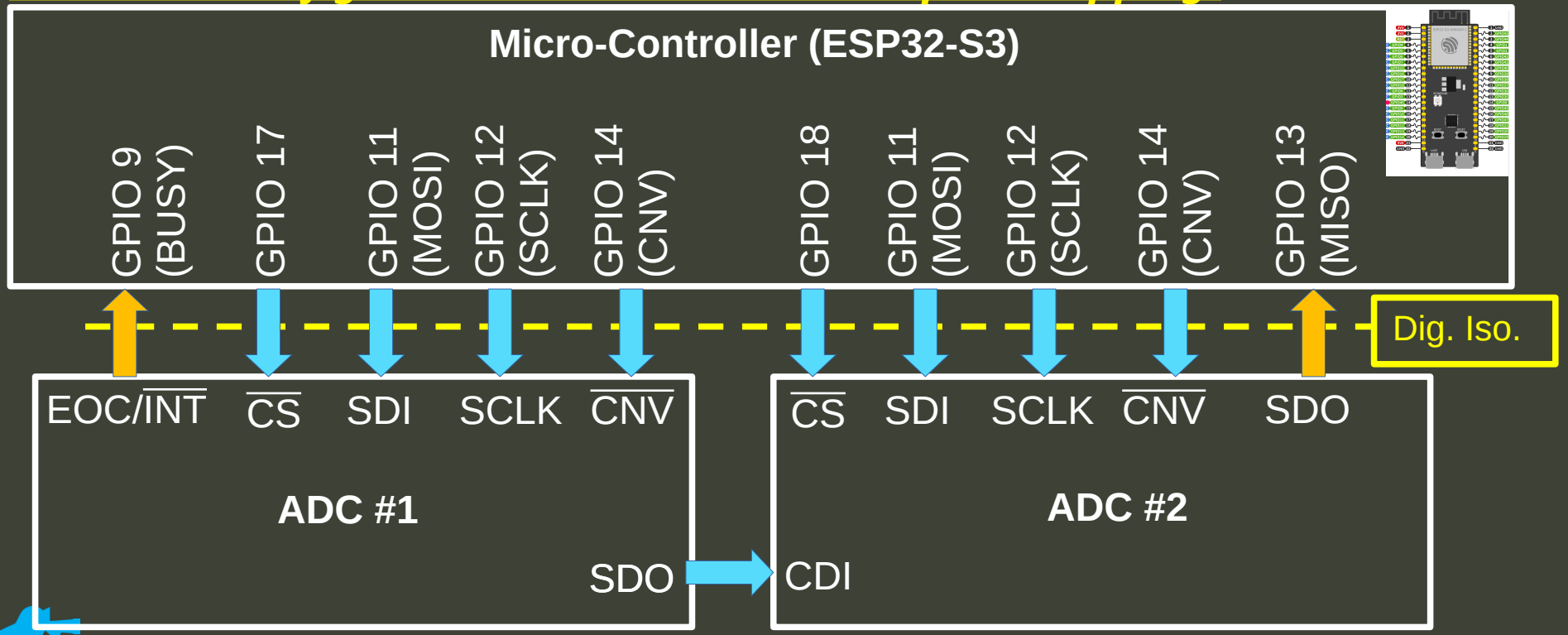


- ***ADS8329 → ADC → Communication Interface:***



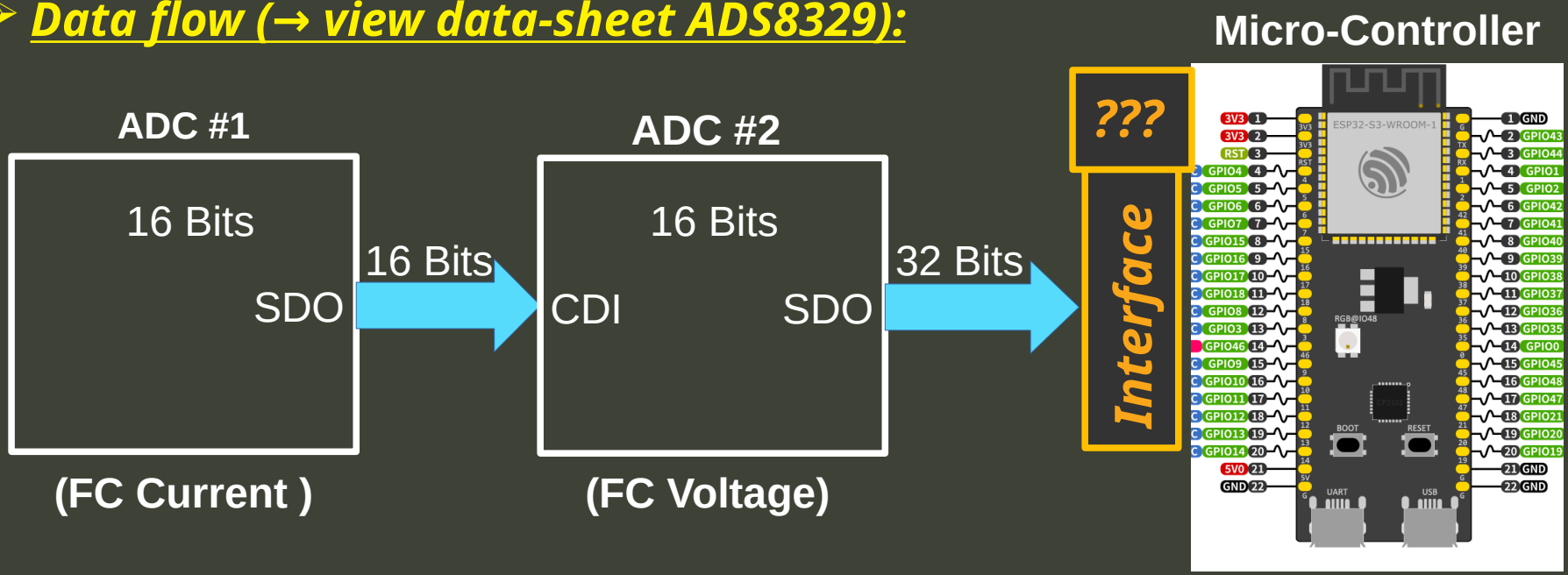
Project Seminar – Board Software Dev.

- ***ADC chain configuration & Micro-Controller port mapping:***



Project Seminar – Board Software Dev.

- ***Data flow (→ view data-sheet ADS8329):***



Project Seminar – Board Software Dev.



- ***ADS8329 → ADC → data-sheet (extraction):***

Chain Mode

The ADS8329/30 can operate as a single converter or in a system with multiple converters. System designers can take advantage of the simple high-speed SPI compatible serial interface by cascading them in a single chain when multiple converters are used. A bit in the CFR is used to reconfigure the EOC/$\overline{\text{INT}}$ status pin as a secondary serial data input, chain data input (CDI), for the conversion result from an upstream converter. This is chain mode operation. A typical connection of three converters is shown in Figure 59.

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- ***ADS8329 → ADC → data-sheet (extraction):***

Chain Mode

The ADS8329/30 can operate as a single converter or in a system with multiple converters. System designers can take advantage of the simple high-speed SPI compatible serial interface by cascading them in a single chain when multiple converters are used. A bit in the CFR is used to reconfigure the EOC/$\overline{\text{INT}}$ status pin as a secondary serial data input, chain data input (CDI), for the conversion result from an upstream converter. This is chain mode operation. A typical connection of three converters is shown in Figure 59.

→ SPI = Serial Peripheral Interface

Project Seminar – Software Dev. - SPI

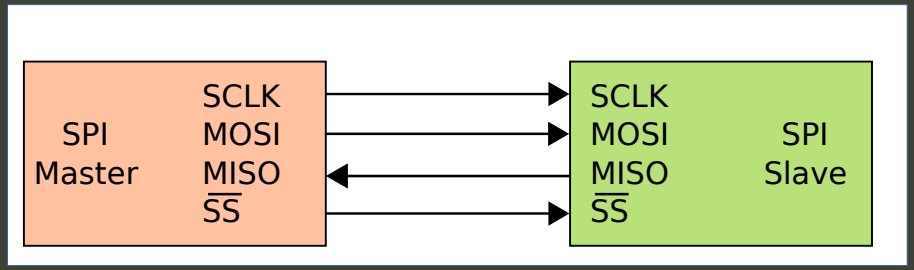




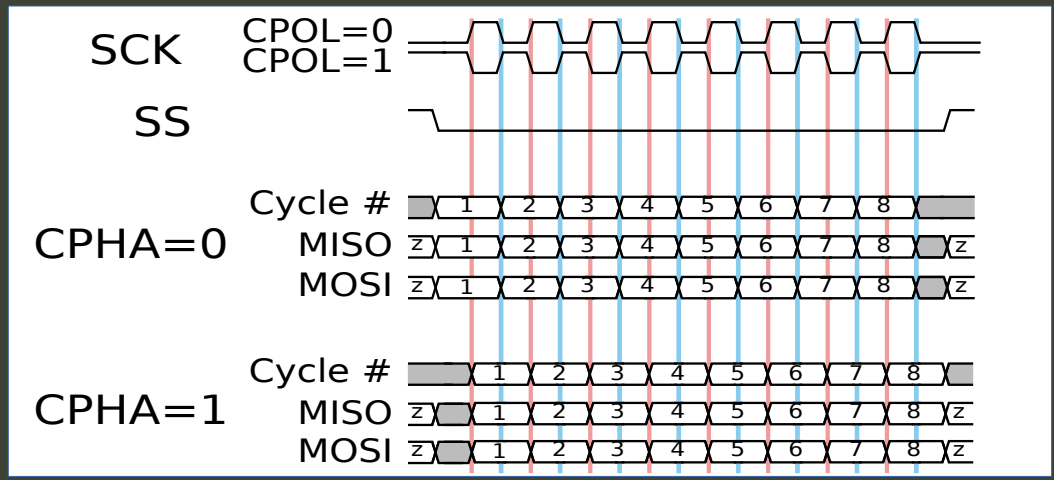
SPI Communication ?

Single Slave Config.

SPI Modes



Source: www.wikipedia.de

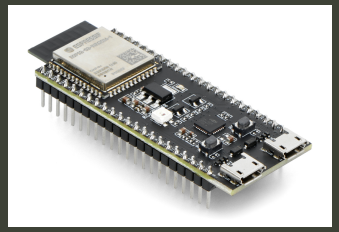


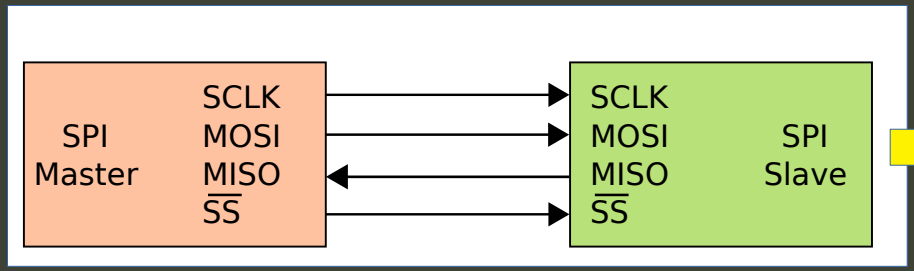
Project Seminar – Software Dev. - SPI



SPI Communication ?

Single Slave Config.

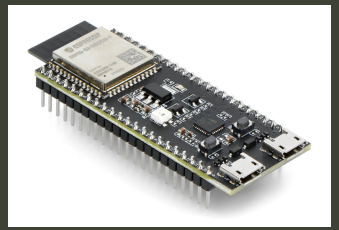




Source: www.wikipedia.de

- ***SCLK → Serial Clock***
- ***SS → Slave Select***
- ***CS → Chip Select***
- ***MOSI → Master Out Slave In***
- ***MISO → Master In Slave Out***

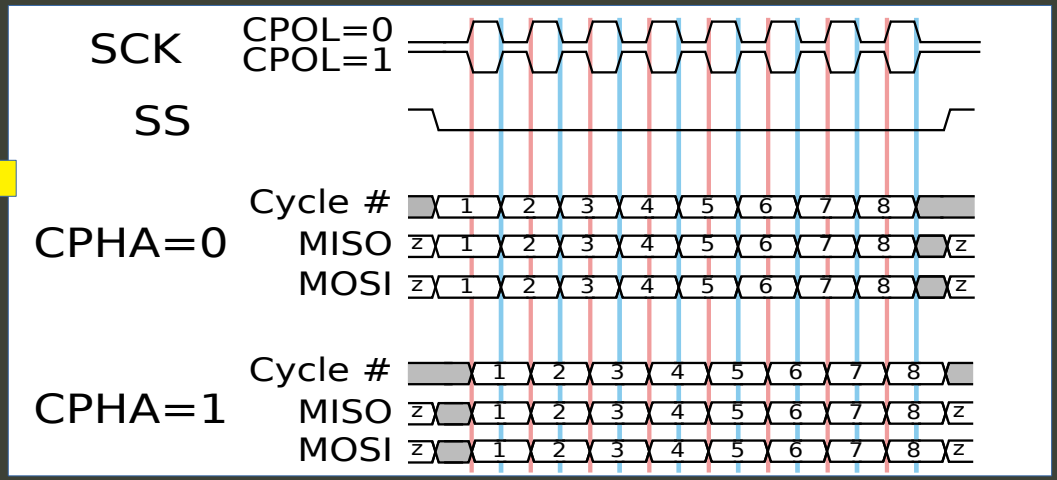
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SPI Communication ?

SPI Modes

- ***SCLK → Serial Clock***
- ***SS → Slave Select***
- ***CPHA → Clock Phase***
- ***CPOL → Clock Polarity***



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- ***ADS8329 → ADC → data-sheet (extraction):***

DIGITAL INTERFACE

The serial clock is designed to accommodate the latest high-speed processors with an SCLK frequency up to 50 MHz. Each cycle is started with the falling edge of FS/$\overline{CS}$. The internal data register content which is made available to the output register at the EOC presented on the SDO output pin at the falling edge of FS/$\overline{CS}$. This is the MSB. Output data are valid at the falling edge of SCLK with $t_{d(SCLKF-SDOVALID)}$ delay so that the host processor can read it at the falling edge. Serial data input is also read at the falling edge of SCLK.

The complete serial I/O cycle starts with the first falling edge of SCLK after the falling edge of FS/CS and ends 16 (see NOTE) falling edges of SCLK later. The serial interface is very flexible. It works with CPOL = 0 , CPHA = 1 or CPOL = 1, CPHA = 0. This means the falling edge of FS/$\overline{CS}$ may fall while SCLK is high. The same relaxation applies to the rising edge of FS/$\overline{CS}$ where SCLK may be high or low as long as the last SCLK falling edge happens before the rising edge of FS/$\overline{CS}$.

→ … we have 2 options:

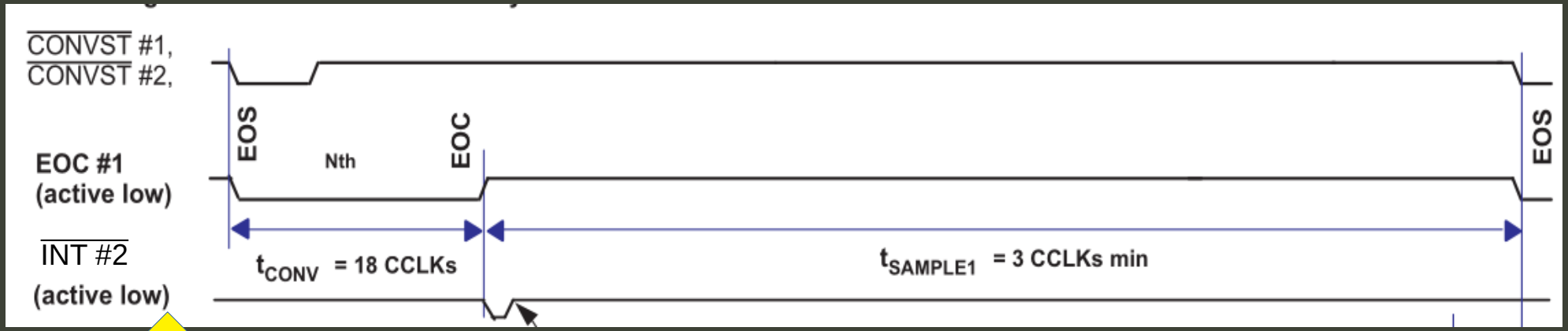
a) CPOL=0 and CPHA=1

b) CPOL=1 and CPHA=0

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- ***ADS8329 → ADC → data-sheet (extraction):***

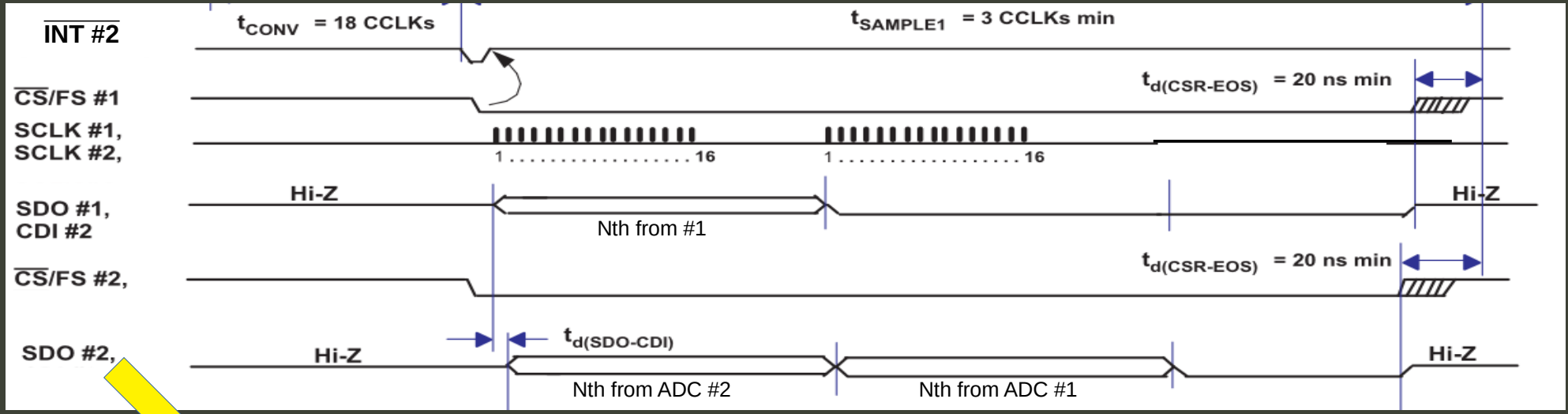


- ***CONVST → Conversion Start***
- ***EOS → End Of Sampling***
- ***EOC → End Of Conversion***
- ***INT → Interrupt***

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- ***ADS8329 → ADC → timing diagram:***



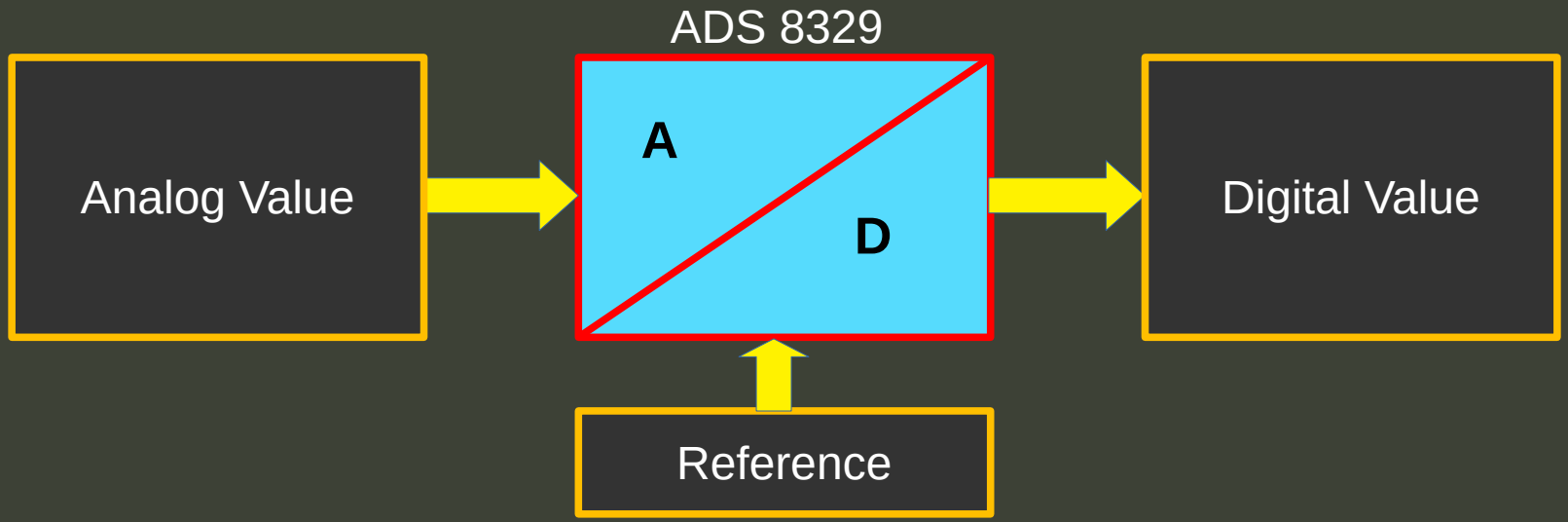
- ***First 16 bit data package belongs to ADC #2***
- ***Second 16 bit data package belongs to ADC #1***

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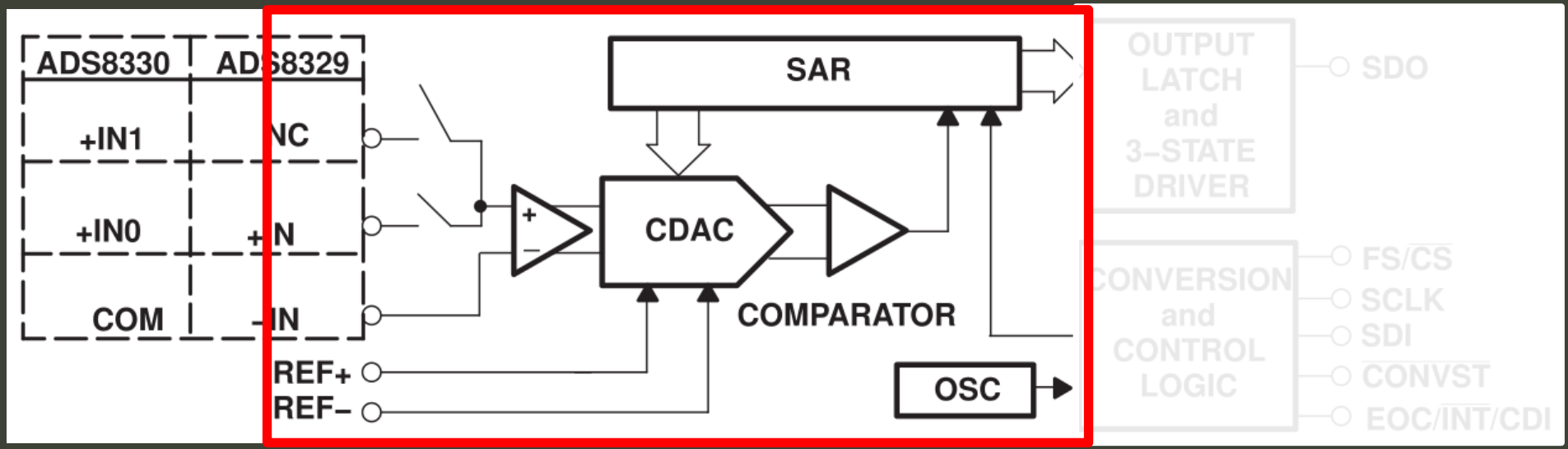
- ***ADS8329 → Analog to Digital Conversion → Example:***





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- ***ADS8329 Block diagram→ Analog to Digital Conversion → Example:***



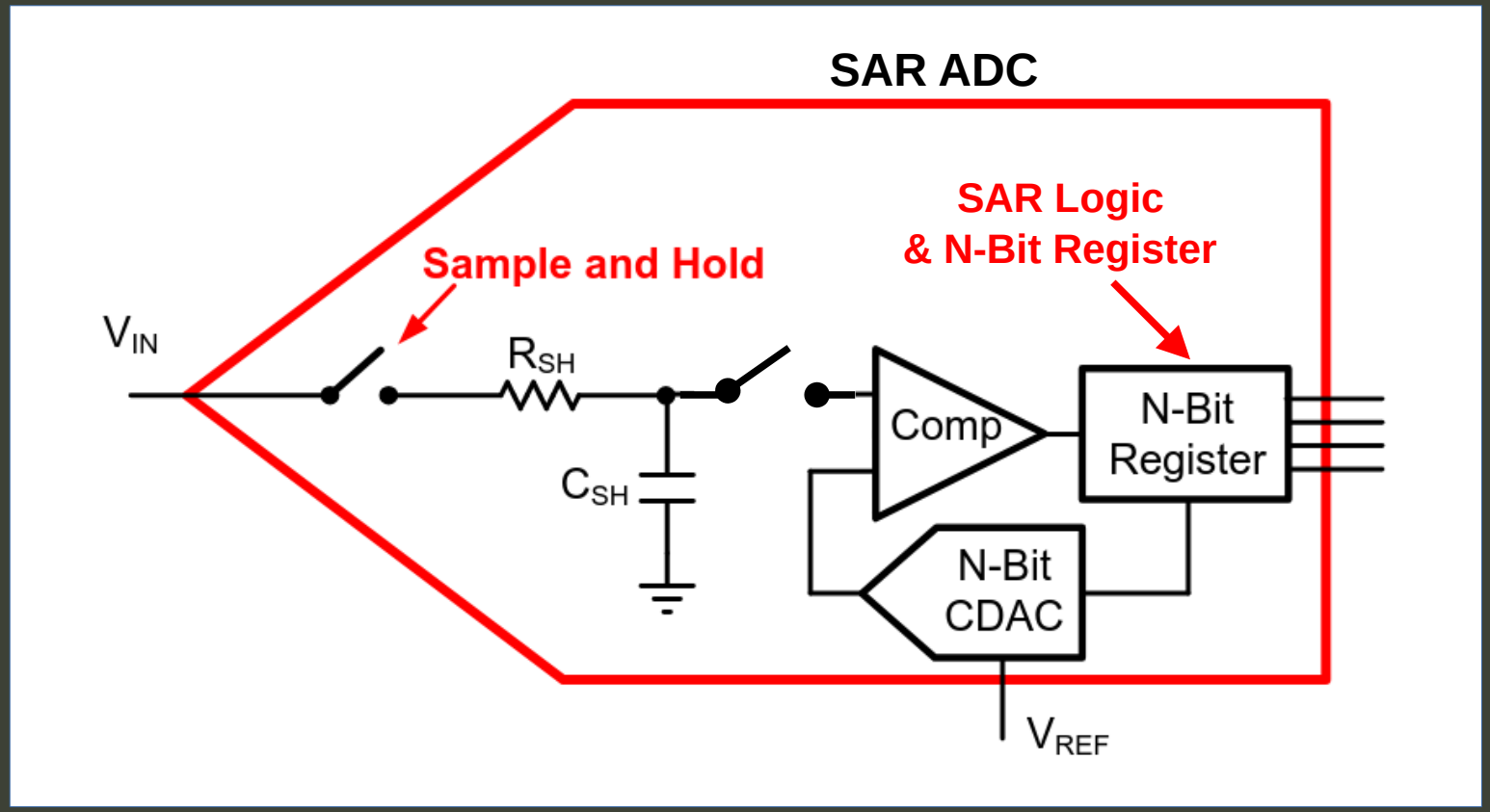
→ SAR = Successive Approximation Register

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- ***SAR ADC → Analog to Digital Conversion → Example:***

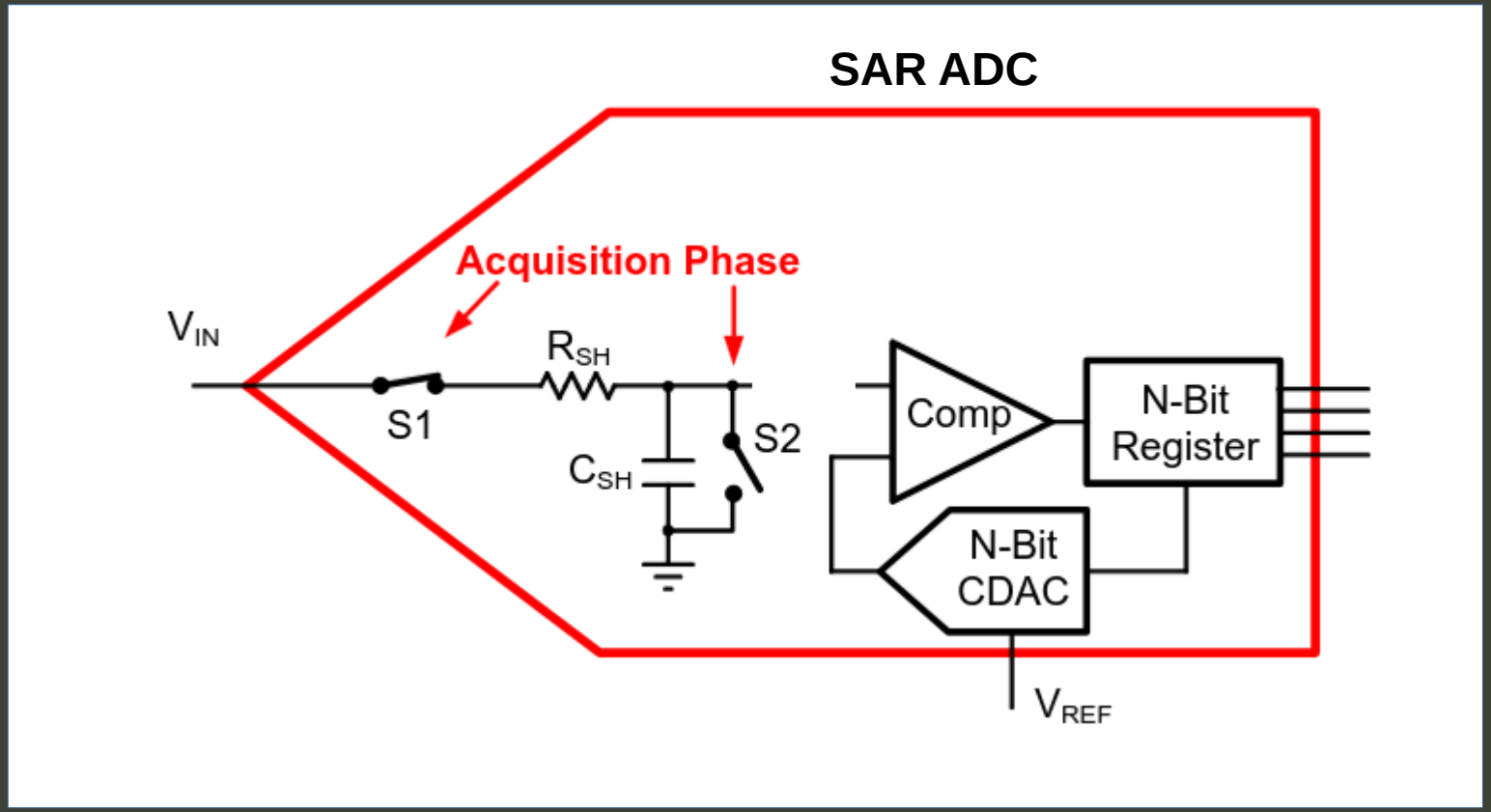


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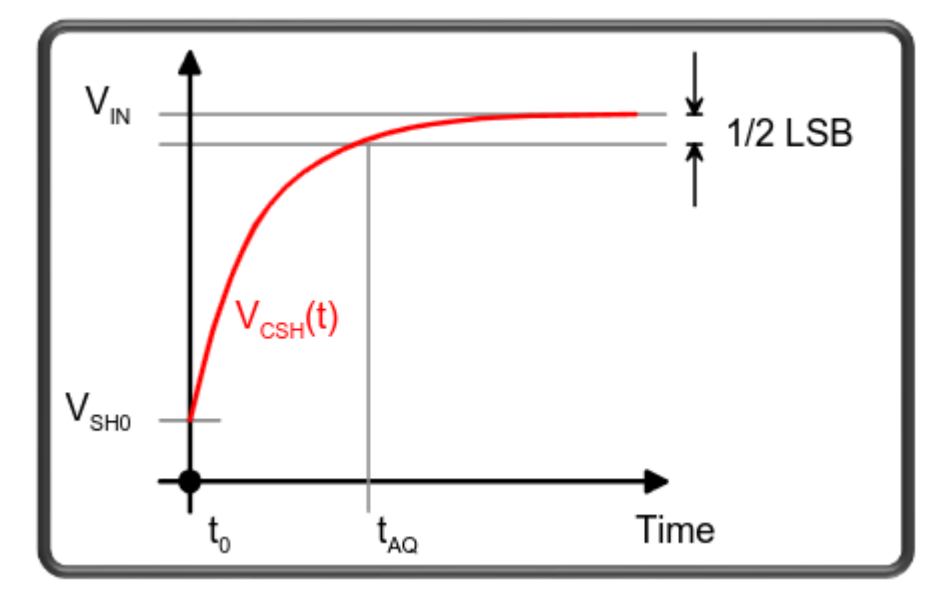


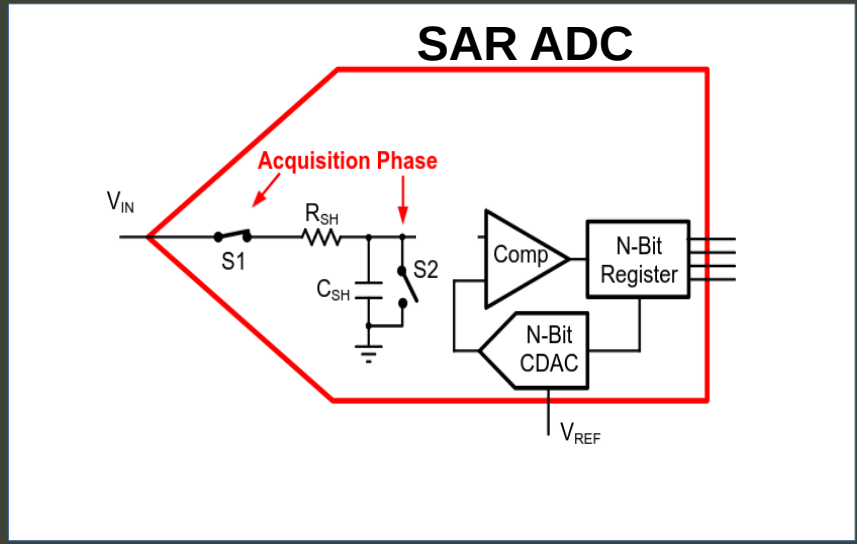
- ***SAR ADC → Analog to Digital Conversion → Example:***



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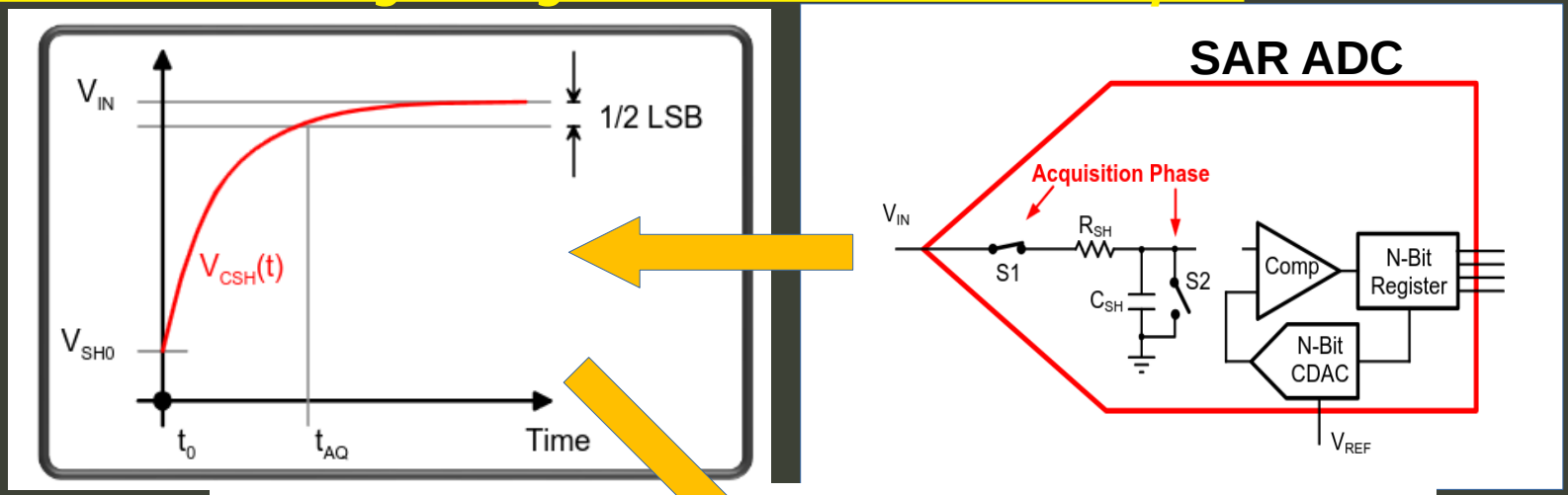
- ***SAR ADC → Analog to Digital Conversion → Example:***





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- ***SAR ADC → Analog to Digital Conversion → Example:***



$$V_{CSH}(t) = V_{CSH}(t_0) + [V_{IN} - V_{CSH}(t_0)] \times (1 - e^{-\frac{t}{\tau}})$$

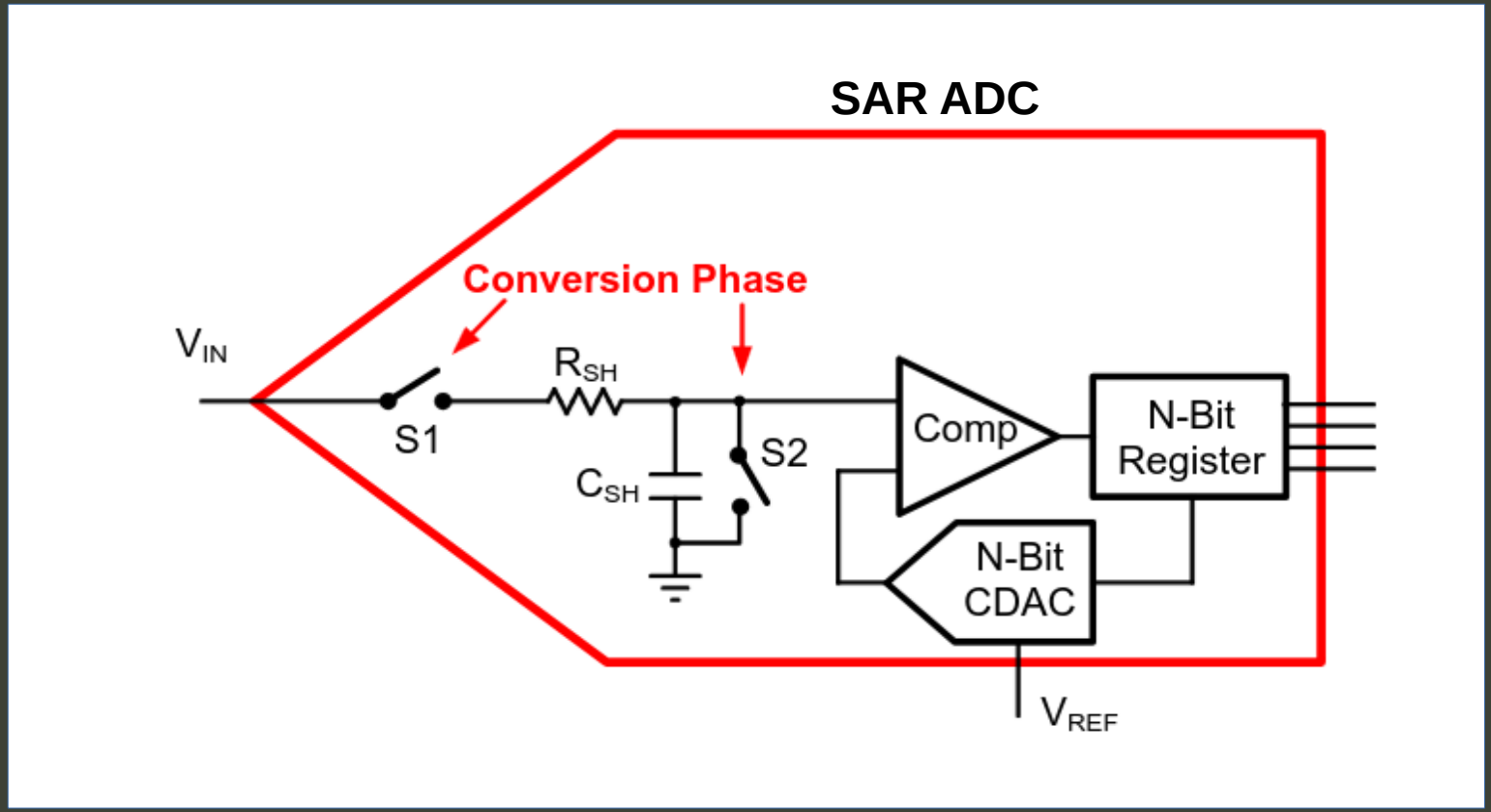
$$\tau = R_{SH} \times C_{SH}$$

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- ***SAR ADC → Analog to Digital Conversion → Example:***

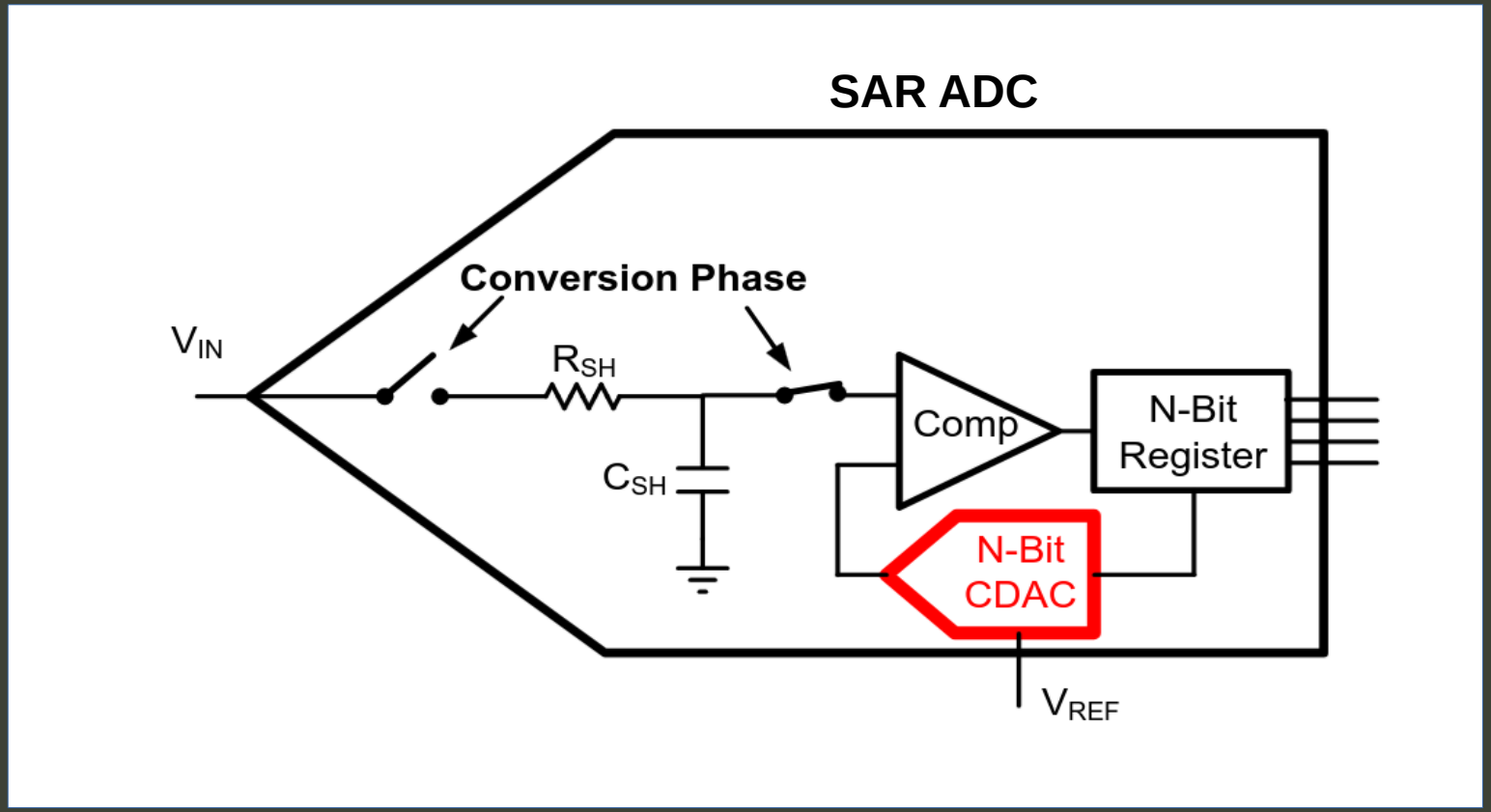


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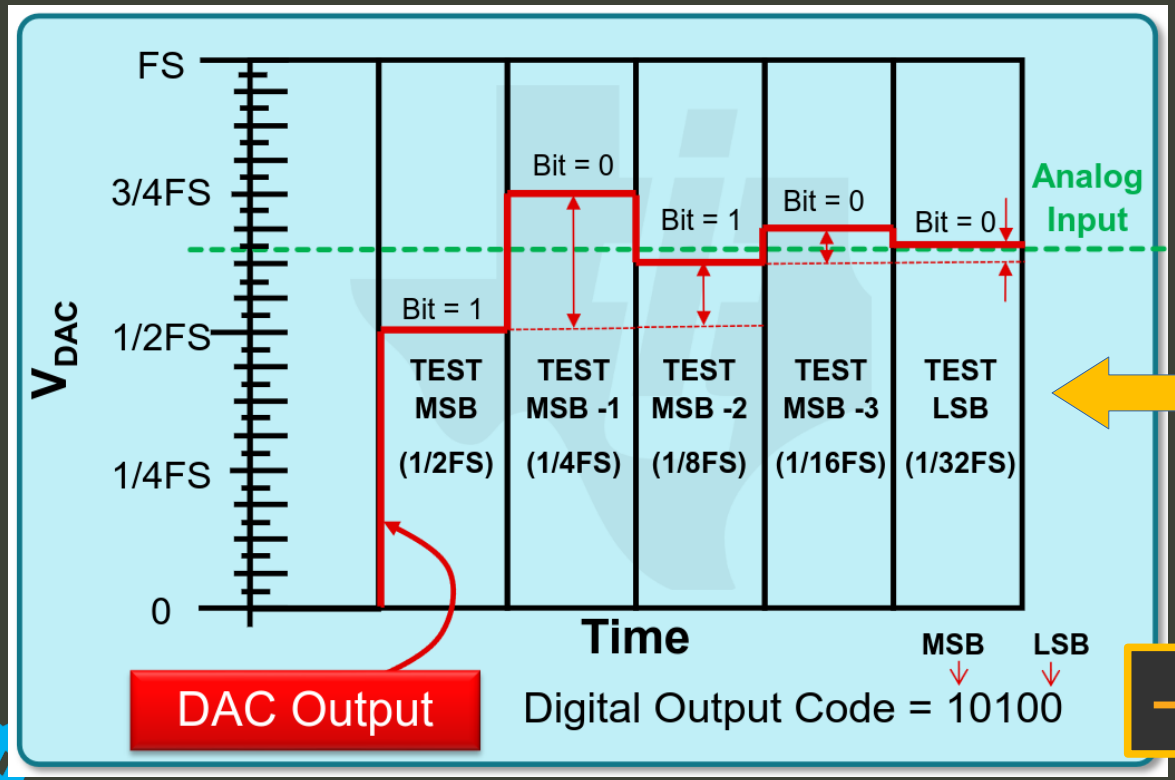


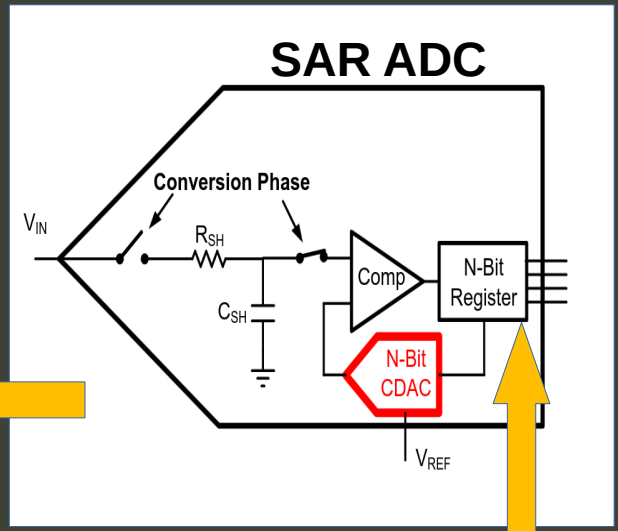
- ***SAR ADC → Analog to Digital Conversion → Example:***



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- ***SAR ADC → Analog to Digital Conversion → Example:***





→***Digital Output Code 10100***

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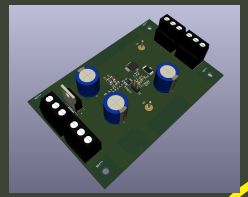




→ *Exercise 10 to SAR ADC is presented in the lecture!*



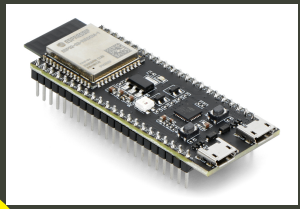
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SPI Communication

ADS8329

ESP32-S3 SPI



SPI:

Serial Peripheral Interface

- Pin description
- Pin functions
- Extract important parameters
- Setting up the SPI-Interface
- Send measurements

- Pin description
- Pin functions
- Extract important parameters
- Setting up the SPI-Interface
- Receive measurements

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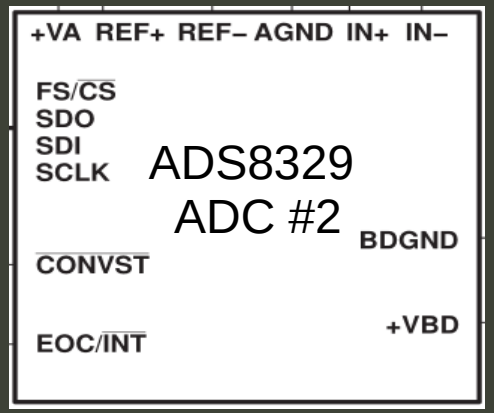


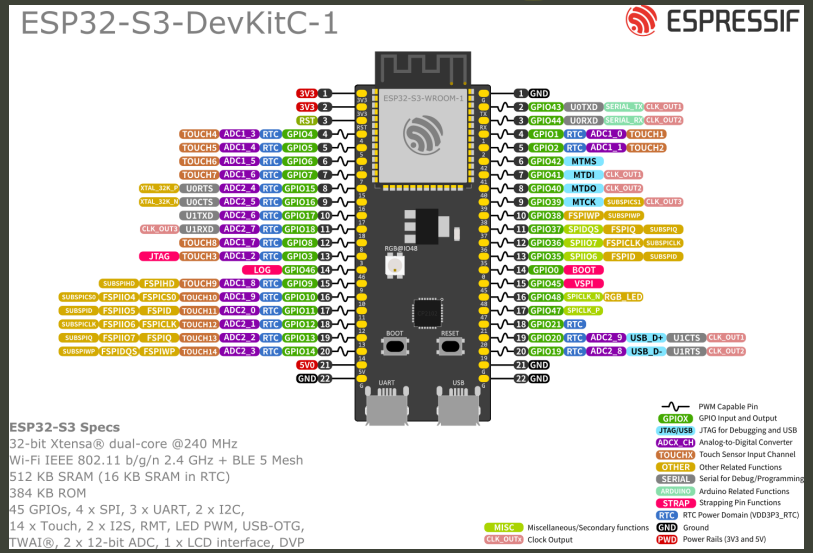


SPI Communication

Slave
- ***ADC***

Master
- ***ESP32-S3***





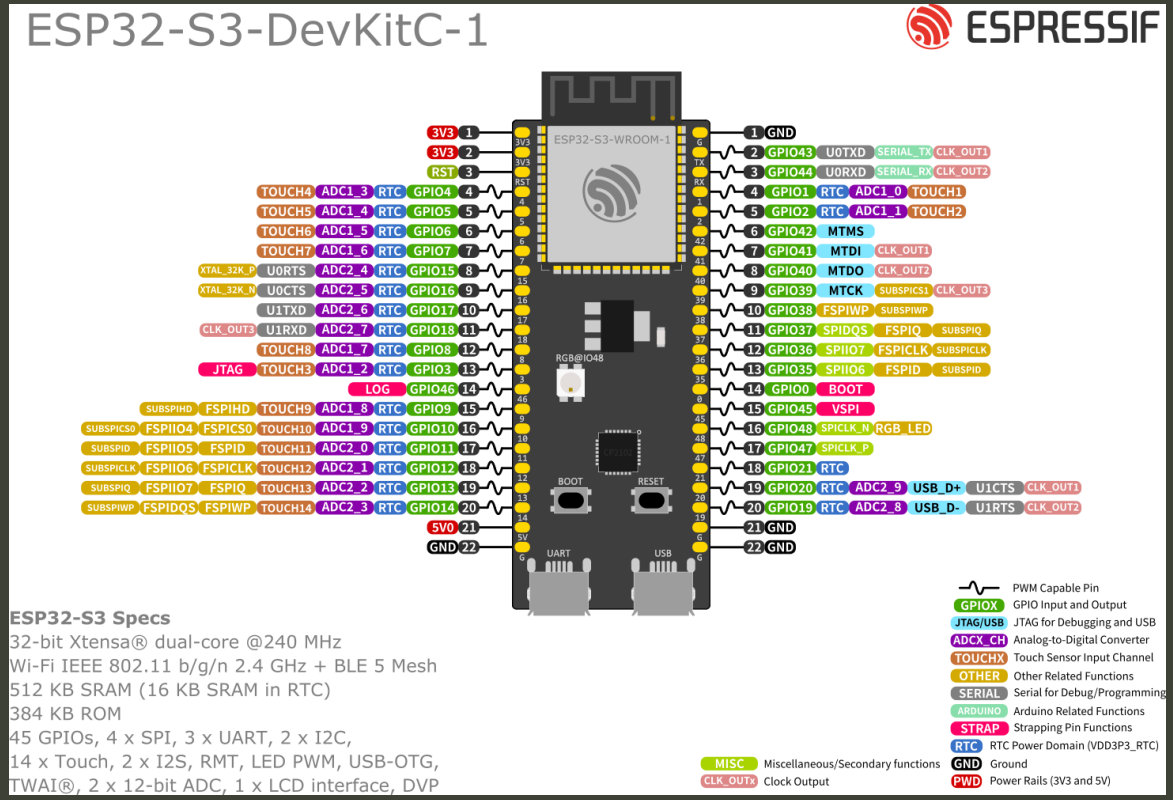
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Master
➢ **ESP32-S3**

- *pins_arduino.h*
- *Busy : GPIO 9*
- *SS : GPIO 10*
- *MOSI : GPIO 11*
- *MISO : GPIO 13*
- *SCK : GPIO 12*
- *$\overline{CS}$: GPIO 18*
- *$\overline{CNV}$: GPIO 14*



Project Seminar - SS2026



Project Seminar - Schedule



- ***Lessons and tasks:***
  - 05.06.2026: independent study – complete PCB design & repetition
  - until 11.06.2026: Exercise 10 → Repeat the exercise 10 - Write down the task of the exercise and the corresponding solution in your report!
  - until 11.06.2026: Write chapter 5 of your project report:
    → Title: “Data transmission via Serial Peripheral Interface - SPI”
    → answer: What do you know about data transmission between an ADC and a micro-controller via SPI? Which parameters must be configured for SPI-communication?

Project Seminar - Schedule



- ***Lessons and tasks:***
  - until 11.06.2026: Write chapter 6 of your project report:
    → Title: “Analog to Digital Conversion via SAR Analog to Digital Converter”
    → answer: What do you know about Analog to Digital Conversion using a SAR Analog to Digital Converter? What functional blocks make up a SAR ADC? How does a SAR ADC convert an analog input voltage into a digital (binary) output voltage? → Do not forget to include the relevant mathematical equations!
- ***next lessons:***
  - 12.06.2026: Lesson 10 – Development of operational board software

… schedule is continously updated!

Project Seminar - SS2026



Thank you !
Have a nice weekend!



